Shape versus circuit: graphene geometric diodes and Tesla valves as terahertz rectifiers

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Geometric diodes, in which an asymmetric channel is meant to rectify by its shape, and electronic Tesla valves have both been proposed as junction-free rectifiers for terahertz rectennas in graphene. We compute the rectification of one set of funnel and Tesla-valve geometries in the ballistic, viscous and plasmonic transport limits with validated solvers, and separate in each what the shape contributes from what the circuit contributes. The Tesla valve does not rectify ballistically, rectifies a steady viscous flow only at Reynolds numbers above the drift-velocity limit of the graphene Fermi liquid, and adds 10^{-5} to 10^{-3} A/W at terahertz frequencies. With the gate included, a steady current in any channel, straight ones included, shows an asymmetry of 0.48 times the bias over the gate overdrive when the polarity is reversed, and less than 10^{-3} when the contacts are exchanged. At terahertz frequencies the gated fluid is plasmonic: a funnel matched to a 75 Ω antenna responds at 0.07 to 0.34 A/W, nearly all of it the Dyakonov-Shur asymmetry of its terminations, and its conversion efficiency follows the square law known for field-effect detectors up to the gate overdrive and peaks below one per cent for every carrier density and gate distance. The ballistic funnel is the one case in which the shape itself rectifies at terahertz frequencies, and its length decides whether it does: with a uniform imposed field the response depends on fL/v_F alone, falls to half near 0.4 and vanishes above 1.7, which limits the 6 μm funnel with the best documented dc asymmetry to tens of gigahertz. For a 0.55 μm funnel we also compute the field, in a particle simulation with a gate, metal contacts and displacement current. At the room-temperature mean free path it rectifies 0.12 A/W per absorbed watt at 1 THz, a sixth of what the imposed fields give, because a quarter of the bias drops at the contacts. The figure is 0.10 A/W at a mean free path of 100 nm, the response is lost with half-diffuse edges, and under a gate at 30 nm a straight channel with its drain tied to the gate rectifies more than the funnel shape. In no regime does the conversion efficiency reach one per cent. A short graphene funnel with specular edges is a square-law detector at the level of the plasmonic element, with a noise-equivalent power of the order of 100 pW/Hz$^{1/2}$ at room temperature; it is not an energy harvester, and the Tesla valve is neither.

I. INTRODUCTION

A rectenna converts radiation to direct current by feeding an antenna into a rectifier. Above a few hundred gigahertz the rectifier is the difficulty: the RC time of a junction diode limits Schottky rectifiers to the low terahertz range, and the search for a junction-free, low-capacitance nonlinearity that works to 1 THz and beyond is long standing. Graphene has attracted two proposals of this kind, both of which rest on the shape of a channel rather than on a junction.

The first is the geometric diode [1–3]: a channel that narrows asymmetrically, typically a funnel, whose dimensions are below the mean free path so that carriers are steered ballistically by the walls. With planar geometry and negligible capacitance it was proposed explicitly for terahertz and infrared rectennas, and the proposal came with a measurement: a 0.5 μm diode with a 75 nm neck in a bow-tie antenna gave a rectified current at zero bias under 28 THz illumination [2]. Encapsulated devices reach reported dc asymmetry ratios of 1.6 to 2 [4, 5]. Related four-terminal ballistic rectifiers have been used for detection and imaging up to 0.7 THz [6, 7] and recently at 3 THz [8], and funnel-shaped rectifiers of chemical vapor deposited graphene respond at zero bias from 0.49 to 0.68 THz [9]; geometric diodes of silicon nanowires have been measured with a flat response to 40 GHz [10]. On the side of theory, a time-dependent particle simulation of a diode of about 100 nm found that its rectification persists into the terahertz range and disappears above about 5 THz, a cut-off the authors place near the Drude damping rate [11], and self-consistent simulations of ballistic funnelling in graphene resonators find nonlinear response from below 10 to tens of terahertz [12]. What these results leave open is what sets the upper frequency of a given geometric diode, and whether the long devices with wide leads, which show the best dc asymmetry, share the speed of the short ones.

The second proposal is the electronic Tesla valve. Tesla's valvular conduit [13] rectifies fluid flow by inertia [14], and in the temperature window where electron-electron collisions make the graphene electron liquid viscous [15, 16] a Tesla-shaped channel has been reported to show a five per cent asymmetry of its four-terminal resistance [17]. A compressible hydrodynamic simulation of a graphene Tesla valve gave a diodicity of 1.07 [18], and in a GaAs electron gas a Tesla valve has reached a diodicity of about 40 above a threshold current [19]. A shape that rectifies an electron fluid with no junction is an obvious rectenna candidate.

What the shape can do is constrained from the outset. For an elastic two-terminal conductor the unitarity of the scattering matrix implies equal transmission in the two directions at every energy, so the linear response is symmetric whatever the shape. Rectification is a second-order effect, and a consistent theory of it requires the re-

sponse of the potential landscape inside the device to the bias [20–23]; an asymmetric lowering of a barrier by the bias has indeed been proposed as the origin of rectification in geometric diodes in place of ballistic steering [24]. The useful question is therefore not whether a shape rectifies but which nonlinearity it feeds, how strong that nonlinearity is at the power an antenna delivers, and how fast it is. A geometric diode in the ballistic regime feeds the bias-induced landscape; a Tesla valve in the viscous regime feeds the convective term of the Navier-Stokes equation; and at terahertz frequencies a gated graphene channel is a plasmonic cavity whose convective and density nonlinearities rectify by the mechanism of Dyakonov and Shur [25, 26], which is the basis of most graphene terahertz detectors to date [27, 28]. For field-effect detectors of this kind the conversion efficiency is known to be quadratic in the amplitude at small signal, to saturate when the amplitude reaches the gate swing [29–31], and to have an absolute maximum of the order of ten per cent, reached below threshold [32]; a silicon transistor rectenna has delivered 13.6% at 263 GHz [33]. These mechanisms have been studied separately, with different geometries and different figures of merit. Whether a geometric shape, as opposed to the terminations of the channel and its connection to the gate, carries the response has not been tested with one set of devices in all of them.

This paper does that. We take one set of geometries (Fig. 1): the funnel of the best-documented geometric diode [4], the classical Tesla valve at the same minimum feature size, a straight control, versions scaled to a 1 μm channel that a 75 Ω antenna can drive, and four shorter funnels down to 0.5 μm. We compute their rectification (i) in a semiclassical ballistic model of the kind used in the geometric-diode literature [2, 34, 35], (ii) in steady viscous electron hydrodynamics, without and with the gate, (iii) in the gated compressible (plasmonic) electron fluid from 10 GHz to 3 THz, at small signal and at finite amplitude up to the gate overdrive, (iv) in the ballistic model driven from 0.3 GHz to 30 THz, and (v), for the shortest funnel, in a particle simulation that computes the field from the charge of the electrons, a gate and the contacts. Each solver is validated against analytic limits and benchmarks before it is used; the acceptance tests are listed in the Supplemental Material.

The answer separates cleanly into shape and circuit, and it differs between the two devices. The Tesla valve does not rectify ballistically, rectifies viscously only at Reynolds numbers the graphene Fermi liquid cannot sustain, and contributes 10^{-5} to 10^{-3} A/W at terahertz frequencies. In the electron fluid the funnel does no better as a shape: with the gate included, a steady current sees a percent-level asymmetry that is the same for every channel and belongs to the gate connection, and the terahertz response of 0.1 to 0.3 A/W is the Dyakonov-Shur asymmetry of the terminations, with a conversion efficiency below one per cent at every operating point for which the fluid description holds. The ballistic funnel is different. Its shape does rectify at terahertz frequencies, on conditions. The first is its length: the response is a function of the reduced frequency fL/v_F and is gone when the flight through the device takes longer than about two periods. The 6 μm device of the dc experiments fails that condition at 1 THz; a 0.55 μm funnel meets it. The second is the field. With an imposed potential, the simplest assumption of simulations of geometric diodes, the short funnel appears to give several times the current per absorbed watt of the plasmonic element; with the potential computed it gives a sixth of that, about what the plasmonic element gives at room temperature. The third is the edges: with half-diffuse walls the response is gone. And when the gate is closer than the device is long, the gate connection again rectifies more than the shape. As a detector the short funnel with specular edges is where the plasmonic element and the best measured graphene detectors are. As a harvester neither element exists: no mechanism studied here converts as much as one per cent of the incident power.

Section II describes the devices and the models, Sec. III the dc results, Sec. IV the plasmonic element from small signal to saturation, Sec. V the ballistic element at terahertz frequencies, its scaling with length and the short funnel with a computed field, and Sec. VI compares the regimes, places the numbers against measured detectors and states the limitations.

II. DEVICES AND MODELS

A. Geometries

All devices are two-terminal planar channels with the source contact on the left and the drain on the right. The funnel [Fig. 1(a)] has the dimensions of device D of Nguyen *et al.* [4]: leads 2 μm wide, a taper at 30 degrees to the axis down to a neck of 175 nm, and an abrupt shoulder on the drain side, 6 μm from end to end, with the contacts on the two ends; its measured two-terminal asymmetry is 1.66 at 80 μA. The classical Tesla valve [Fig. 1(b)] is built from channel centre lines: a zig-zag main channel of width 175 nm inclined at ±25 degrees, and at every vertex a loop that continues the arriving segment straight on, turns through a half circle of radius 175 nm and rejoins the main channel against the stream. Flow from left to right follows the zig-zag; flow from right to left is split into the four loops. We also keep three earlier variants in which the loops are attached to one side of a straight channel, because they illustrate that the conclusion does not depend on the details of the valve, and a straight 175 nm channel between the same leads as a control. For the terahertz calculations each shape is also scaled so that its narrowest width is 1 μm, with 4 μm contacts, which brings the device resistance from kilohms to the hundred-ohm range of an antenna [Fig. 1(c)]. Four further funnels, from a half-scale copy of device D down to a 0.5 μm device with the proportions of the diode

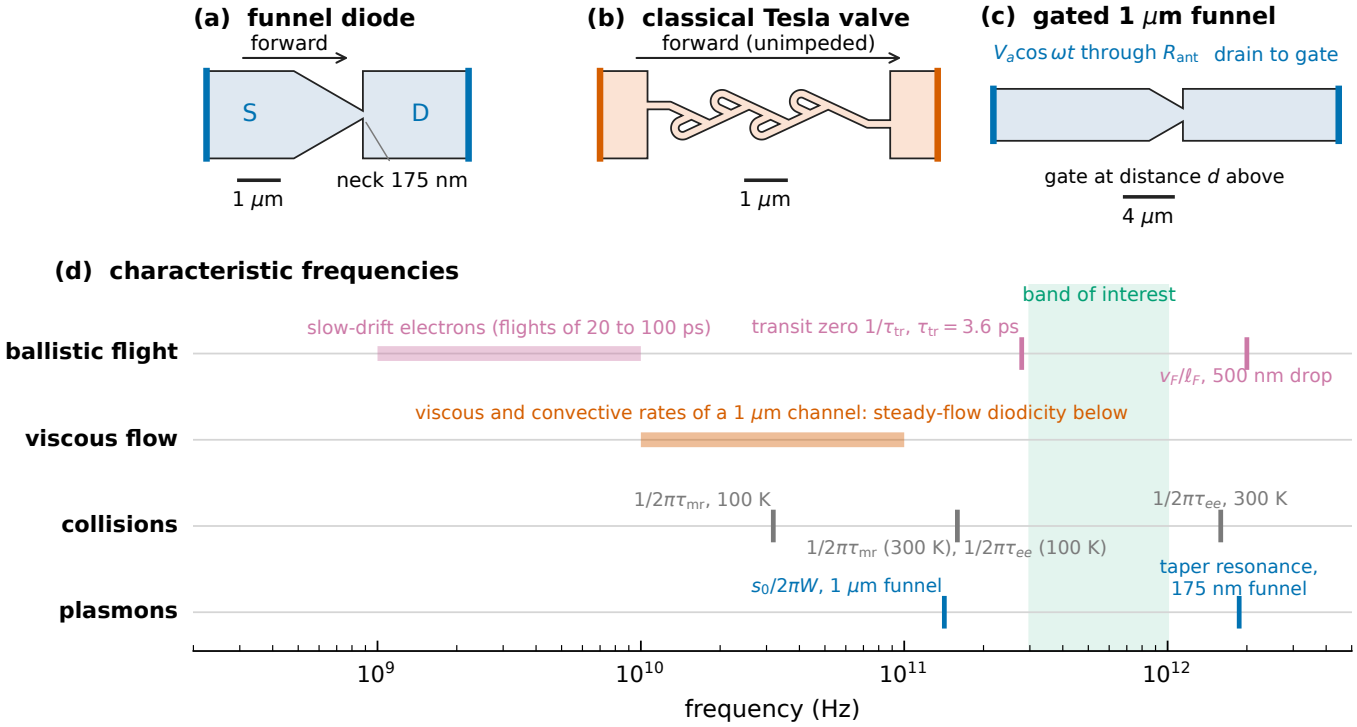


FIG. 1. Devices and frequency scales. (a) The funnel geometric diode, device D of Ref. [4]: 2 μm leads, a 30 degree taper to a 175 nm neck and an abrupt shoulder; forward is the direction into the taper. (b) The classical four-loop Tesla valve at the same 175 nm channel width. (c) The gated funnel scaled to a 1 μm neck with 4 μm contacts, with the terminations used for the plasmonic response: the antenna, a Thevenin source of resistance $R_{\rm ant}$, drives the source contact and the drain is tied to the gate. (d) Characteristic frequencies of the mechanisms studied, for the parameters of Table I; the shaded band is the range of interest for a terahertz rectenna. The shorter funnels of Sec. V C are drawn in Fig. 9(a).

measured at 28 THz [2], are used for the length scaling of Sec. V C and listed in Table II.

B. Ballistic model

The ballistic model is the semiclassical trajectory (Monte Carlo [36]) method of the geometric-diode literature [2, 34, 35]. Electrons with the Dirac dispersion $E = \hbar v_F |\boldsymbol{k}|$, $v_F = 10^6$ m/s, are injected at the contacts with the flux-weighted cosine law, fly along classical trajectories, reflect from the walls specularly with probability p and diffusely otherwise, and suffer isotropic momentum-relaxing collisions after exponentially distributed free paths of mean λ. The current is the Landauer expression

$$I = \frac{ge}{h} \int dE\, N(E)\, T(E;V) \left[f_S(E) - f_D(E) \right], \qquad (1)$$

with $g = 4$, $N(E) = EW/\pi\hbar v_F$ the number of modes of the injecting contact of width W, T the transmission probability estimated from the trajectories, and $f_{S,D}$ the Fermi functions of the contacts at electrochemical potentials E_F and $E_F - eV$. Energies are sampled from the bias window $N|f_S - f_D|$, so the estimator has zero variance at zero bias.

With no potential inside the device, flux conservation gives $N_S T_{SD} = N_D T_{DS}$ at every energy and Eq. (1) is odd in V for any shape; the trajectories cannot rectify. Rectification enters through the potential energy $U(\boldsymbol{r}; V)$ that the bias creates inside the channel [20]. We use the simplest assumption of the published simulations, a uniform field along the channel, $U = -eVx/L$ over the device length L, and integrate the curved flights in chords with a controlled sagitta. Self-consistent electrostatics is known to roughly halve the asymmetry of such diodes relative to the uniform-field value and to reverse it for shallow tapers [35]; we return to the field distribution in Sec. V. The figure of merit is the asymmetry $A = I(+V)/|I(-V)|$; writing $I = G_1 V + G_2 V^2$, $A \simeq 1 + 2(G_2/G_1)V$, and a cosine drive of amplitude V_a rectifies $I_{\rm dc} = G_2 V_a^2/2$ if the response is instantaneous. Carriers are electrons at density $n = 1.4 \times 10^{12}\,{\rm cm}^{-2}$ ($E_F = 0.138$ eV) and 300 K, the density we estimate from the gate voltage and lever arm reported for that device [4]; holes give the inverse asymmetry. The model has no length of its own: with the bias, E_F and $k_B T$ fixed, the transmission depends only on the shape and on λ/L, and under a drive of frequency f only on those and on fL/v_F, while currents and conductances scale with the width. This is used, and tested, in Sec. V C.

The solver was validated on eleven tests before

use: unit transmission of a specular channel, the free-molecular transmission factors of a diffuse channel [37], the Sharvin conductance [38], the ballistic-to-diffusive crossover, reciprocity with and without the field, unit asymmetry of a mirror-symmetric constriction, and convergence of the chord integration. Classical trajectories require features well above the Fermi wavelength, 30 nm here.

C. Ballistic model with a computed field

For the shortest funnel the potential is computed instead of imposed, in a particle simulation of the kind applied to geometric diodes in Refs. [11, 34]. The graphene sheet lies in a uniform dielectric of permittivity 3.8 at a distance d above a grounded gate plane; d is 30 nm, the gate of the plasmonic model below, or 300 nm, an oxide back gate that leaves a device of this size nearly ungated. The contacts are metal strips in the plane of the sheet, 200 nm long and as wide as the leads, held at potentials $V_S(t)$ and $V_D(t)$. All electrons of the device, about 2900, are represented by 4 to 8×10^6 sample particles that move together, with the trajectories, wall reflection, scattering and equilibrium injection at the contacts of the model above. At every time step of 2.5 fs their charge is assigned to a grid of 5 nm spacing and the potential energy

$$U(\boldsymbol{r}) = \frac{e^2}{4\pi\epsilon}\sum_{\boldsymbol{r}'}\delta N(\boldsymbol{r}')\left[\frac{1}{|\boldsymbol{r}-\boldsymbol{r}'|} - \frac{1}{\sqrt{|\boldsymbol{r}-\boldsymbol{r}'|^2+4d^2}}\right] \tag{2}$$

is evaluated by fast Fourier transform; δN is the electron number at a grid node in excess of the uniform background that makes the undriven sheet neutral, the second term is its image in the gate, and the sum includes the charge on the contacts, which is solved for at every step so that the strips are equipotentials. The particles are then advanced in the gradient of the interpolated potential. This is the electrostatics of a thin sheet in three dimensions; it goes over to the local gate capacitance of the plasmonic model for small d and to the ungated sheet for large d, and it contains the plasma waves of the collisionless gas.

The currents at the source, the drain and the gate include the displacement current, so the admittance Y_1 at the drive frequency and the absorbed power $P_{\rm abs} = \langle V_S I_S + V_D I_D\rangle$ are results of the calculation. With the gate present the terminations matter, as in the plasmonic model, and two are used: a drive that leaves only the shape, $V_S = -V_D = (V_a/2)\cos\omega t$, under which a mirror-symmetric device gives no rectified current by symmetry, and the drain tied to the gate, $V_S = V_a\cos\omega t$ and $V_D = 0$, under which a straight channel already rectifies.

Nine tests were passed before use (Supplemental Material), among them the capacitance of the contact strips, the static screening of a weak potential, the speed of the gated plasma wave, which is 0.98 to 0.99 of the collisionless value of $3.90\,v_F$ under the 30 nm gate, the results of the imposed-field model for the same device when the interaction is switched off, and a linear conductance that the interaction leaves unchanged within 1%. The sample particles collide with each other through the grid, which relaxes the Fermi distribution towards a classical one; in the open device a particle is replaced by a contact within about a picosecond, and the rectified current at the operating point of Sec. V E does not change between 4 and 16×10^6 particles, while it is too large by up to 40% with 10^6.

D. Viscous model

Where electron-electron collisions are the fastest scattering process the electron liquid obeys the Navier-Stokes equation [15, 39]. The dc model is steady incompressible flow with momentum relaxation,

$$\mathrm{Re}\,(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + \nabla^2\boldsymbol{u} - \mathrm{Da}\,\boldsymbol{u}, \qquad \nabla\cdot\boldsymbol{u} = 0, \tag{3}$$

in units of the lead width W and the mean inlet speed, with

$$\mathrm{Re} = \frac{I}{ne\nu}, \qquad \mathrm{Da} = \frac{W^2}{\nu\tau_{\rm mr}}, \qquad S = \frac{\ell_s}{W} \tag{4}$$

the Reynolds number, the damping (Gurzhi) number and the slip length ℓ_s of the Navier wall condition [40] in units of W. In two dimensions the Reynolds number depends on the current and not on the width. The device is current biased with a developed inlet profile, and the figure of merit is the diodicity $D = \Pi_r/\Pi_f$, the ratio of the pressure (voltage) drops for flow from drain to source and from source to drain. Equations (3) are solved with Taylor-Hood elements [41] on triangular meshes [42] with Newton iteration and a sparse direct solver [43]; the wall condition $\boldsymbol{u}\cdot\boldsymbol{n} = 0$ is imposed with length-weighted normals [44], which makes the discrete wall flux vanish exactly and removes a spurious diodicity of 10^{-5} that polygonal walls otherwise produce. Validation covers Poiseuille and Brinkman flow with slip, Stokes reciprocity ($D - 1 < 10^{-10}$ at $\mathrm{Re} = 0$), the Kovasznay solution [45], the backward-facing step [46, 47], and the diodicity of the valve itself as a classical fluid device. The discretisation floor of $D - 1$ is 3×10^{-5}.

E. Plasmonic model

At terahertz frequencies the fluid is compressible: a gate at distance d makes the local density follow the local electrochemical potential, and density waves propagate as plasmons. With the gradual-channel approximation and degenerate Dirac carriers the electrochemical potential is $\Phi(n) = e^2n/C_g + \hbar v_F\sqrt{\pi n}$, the hydrodynamic mass is $m^*(n) = \hbar\sqrt{\pi n}/v_F$ [48, 49], and because both depend on the density only the force per unit mass is the gradient

of a single function $h(n)$ with $h' = \Phi'/m^*$. The model is

$$\partial_t n + \nabla \cdot (n\boldsymbol{u}) = 0, \tag{5}$$

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} + \frac{\boldsymbol{u}}{\tau_{\rm mr}} = -\nabla h(n) + \nu\nabla^2\boldsymbol{u}, \tag{6}$$

with the plasma speed $s(n)$ given by

$$s^2(n) = n\,h'(n) = \frac{e^2 n}{C_g m^*(n)} + \frac{v_F^2}{2}. \tag{7}$$

The second term is the quantum capacitance [50]; its share $q = v_F^2/2s_0^2$ of $s_0^2 = s^2(n_0)$ is 0.04 at $n_0 = 10^{12}\,\mathrm{cm}^{-2}$ under 30 nm of hBN and grows to 0.5 at the thin-gate, low-density corner of Sec. IV C. Replacing h by $s_0^2 n/n_0$ gives the shallow-water model of Dyakonov and Shur [25]. At frequencies above $1/\tau_{ee}$ the viscosity is taken in the viscoelastic form $\nu/(1 + i\omega\tau_{ee})$.

The antenna is a Thevenin source of amplitude V_a and resistance $R_{\rm ant}$ connected to lumped ports: each contact is an equipotential whose voltage and current satisfy Kirchhoff's law with the external circuit. Two terminations are used throughout. In the *symmetric* one the antenna drives the two contacts in antiphase through $R_{\rm ant}/2$ each and the gate sits at the midpoint; the circuit is then mirror symmetric and any dc response is due to the shape of the channel. In the *Dyakonov-Shur* (DS) termination the antenna drives the source through $R_{\rm ant}$ and the drain is tied to the gate; the terminations are then asymmetric and a uniform channel already rectifies [26]. The responsivity is the short-circuit dc current per available power,

$$R_I = \frac{I_{\rm dc}}{P_{\rm av}}, \qquad P_{\rm av} = \frac{V_a^2}{8R_{\rm ant}}, \tag{8}$$

and the Johnson-noise-limited noise-equivalent power is $\mathrm{NEP} = \sqrt{4k_BT/R_{\rm dc}}/R_I$ with $R_{\rm dc}$ the dc resistance of the device.

Two solvers are used. For small signals the fields are expanded to second order about rest: the first order is a complex time-harmonic plasmon problem, and the time average of the second order is a Stokes-Brinkman problem for the dc flow, forced by products of first-order fields, whose port current or voltage is the rectified signal (Appendix A). For finite amplitudes Eqs. (5) and (6) are integrated in time with the second-order backward difference formula and Newton iteration, with a bias tee that separates the dc load from the antenna. Both use Taylor-Hood elements on the meshes of the viscous model. The steady states of Eqs. (5) and (6) under a dc current, used in Sec. III C, are obtained with the same discretisation by Newton iteration. The validation ladder has twenty tests: the static limit against the viscous solver, the plasmon dispersion, the Dyakonov-Shur closed form [Fig. 5(a)], reciprocity of the symmetric termination, mesh and time-step convergence, agreement of the two solvers at small amplitude, invariance under co-scaling of the physical parameters at fixed nondimensional groups, and, for the steady states, a closed form for the gated channel and the incompressible limit against the viscous solver.

TABLE I. Parameter sets. The ballistic model follows the experiment of Ref. [4]; the two plasmonic sets are representative of hBN-encapsulated graphene at the two temperatures and are labelled by them; they are assumptions, not fits. The 100 K viscosity is of the order measured [15, 16]; the 300 K value is the kinetic estimate $v_F^2\tau_{ee}/4$. W is the contact width.

Quantity	Value
Ballistic	
n, E_F, T	$1.4\times10^{12}\,\mathrm{cm}^{-2}$, 0.138 eV, 300 K
specularity p	1, 0.5, 0
mean free path λ	∞, 700, 200, 100 nm
Viscous	
current at $\mathrm{Re}=1$	160 μA ($\nu = 0.1$ m^2/s), 32 μA (0.02)
Da, S	0 to 128; 0 to ∞
Plasmonic	
n_0, d, ϵ_r	$10^{12}\,\mathrm{cm}^{-2}$, 30 nm, 3.8
s_0, gate overdrive	$3.6\,v_F$, 1.43 V
"100 K": $\tau_{\rm mr}$, τ_{ee}, ν	5 ps, 1 ps, 0.1 m^2/s
"300 K": $\tau_{\rm mr}$, τ_{ee}, ν	1 ps, 0.1 ps, 0.03 m^2/s
$W/s_0\tau_{\rm mr}$ ($W = 4\,\mu$m)	0.22 (100 K), 1.12 (300 K)
antenna $R_{\rm ant}$	75 Ω

III. RECTIFICATION AT DC

A. Ballistic: the funnel rectifies, the Tesla valve does not

Figure 2 shows the asymmetry at 0.1 V and the resistance of all six devices for three edge specularities and three mean free paths. In the clean specular limit the funnel gives $A = 1.62 \pm 0.02$ (1.25 at 0.05 V), close to the 1.66 measured at 80 μA [4]; the agreement is conditional, because at the mean free path estimated for these devices [4], 700 nm, the same model gives 1.09, and 1.06 with fully diffuse edges. The assumed field matters as much as the scattering. If the same bias is taken to drop over the taper and neck alone, with the leads cut to 250 nm (device G2 of Table II), the asymmetry at $\lambda = 700$ nm is 1.55 and the clean value 1.91; if it drops over 500 nm centred on the neck of the full device it is 0.96, the reverse sign. The experiment has its contacts at the ends of the 6 μm shape and is a two-terminal measurement, so none of these profiles is excluded by its geometry; self-consistent electrostatics is what would decide between them [35]. The choice of field profile, λ and p therefore moves $A - 1$ by a factor of several, and we do not claim a quantitative fit. What the comparison establishes is the mechanism and the scale: electrons accelerated toward the neck by the bias arrive with more kinetic energy and more transverse modes than those decelerated toward it.

No Tesla geometry rectifies in this model. All 54 points of the three side-loop variants lie between 0.93 and 1.08 with standard errors of 0.01 to 0.05, and sixteen of the eighteen points of the classical valve are within two standard errors of one. The two exceptions are in the clean specular limit, where the classical valve gives

$A = 0.93 \pm 0.02$ at 0.05 V and 0.89 ± 0.01 at 0.1 V: the wrong sign, the blocking direction being the easier one, and gone as soon as edges are partly diffuse or bulk scattering is present. The straight control gives 0.99 ± 0.01. Meanwhile the loops cost resistance in both directions, 1.4 to 22 kΩ against 0.5 to 2 kΩ for the funnel [Fig. 2(b)]. The result follows from the structure of the problem rather than from the parameters: at fixed potential landscape the transmission is reciprocal for any shape, and the Tesla loops do nothing to the way the bias landscape acts on the mode count at a constriction, which is what the funnel exploits. It agrees with the experiment, which reports that the rectification of the valve disappears outside the viscous regime [17].

B. Viscous: the Tesla valve works, beyond the reach of the Fermi liquid

In the electron fluid the picture reverses in principle and not in practice. Figure 3(a) shows the diodicity of the 175 nm classical valve and of the funnel against Reynolds number without damping. The valve is a genuine Tesla valve: with slip walls $D-1$ is 0.15 to 0.23 at $\mathrm{Re} = 10$ and 1.1 to 1.5 at $\mathrm{Re} = 30$, growing close to Re^2, the signature of an inertial loss present in one direction only. The fields [Fig. 3(c) and 3(d)] show its origin: in the forward direction the loops are stagnant, in the reverse direction each carries a stream that re-enters the main channel against the flow. As a check on the geometry, the same valve treated as a classical no-slip fluid device gives $D = 1.05$ at $\mathrm{Re} = 50$ and 1.23 at $\mathrm{Re} = 100$; three-dimensional simulations of multistage valves give 1.02 to 1.03 and 1.10 to 1.15 at the same Reynolds numbers [51], a duct of finite depth adding wall friction that the two-dimensional flow lacks. The diodicity of the valve is converged with respect to the mesh to about 10% (Supplemental Material), which is immaterial for what follows. The funnel is a diode of the opposite sign, the forward direction being the harder one: its asymmetry is that of the exits, a jet from the abrupt shoulder against a diffuser along the taper, and it is the stronger of the two below $\mathrm{Re} \approx 3$ ($|D-1| = 1.2 \times 10^{-2}$ at $\mathrm{Re} = 3$ with slip) but saturates near 0.15. This is the convective (Bernoulli) nonlinearity proposed for wedge-shaped channels as a signature of electron hydrodynamics [52] and recently measured in Venturi-shaped GaAs devices [53]; with equal lead widths on the two sides, as here, it is quadratic in the current.

Three facts decide the matter. First, the experiment of Ref. [17] used currents of at most a few microamperes (6.3 μA for the device that showed the effect), for which we estimate $\mathrm{Re} \approx 0.04$ to 0.2 at $10^{12}\,\mathrm{cm}^{-2}$ and $\nu = 0.1$ to 0.02 m^2/s; the authors quote no Reynolds number. For Re up to 0.3 every geometry, wall condition and damping gives $|D-1| < 3 \times 10^{-5}$, three orders of magnitude below the reported five per cent, which was observed for holes in a single-loop valve as a ratio of four-terminal resistances. Steady incompressible electron hydrodynamics does not reproduce that observation; whatever produces it is not the inertial Tesla mechanism, and we do not explain it. The candidates that Eq. (3) excludes are the compressibility of the gated fluid, taken up in Sec. III C, and electron heating. Second, the Reynolds number is bounded. The drift speed in a channel of width w is $u = \mathrm{Re}\,\nu/w$, and keeping it below $0.3\,v_F$, where a Fermi-liquid description is still tenable, caps Re at 0.5 to 2.6 in a 175 nm channel and at 3 to 15 in a 1 μm channel for $\nu = 0.1$ to $0.02\ \mathrm{m}^2/\mathrm{s}$ [shaded in Fig. 3]. The bound is not new: the Reynolds number of the graphene electron liquid scales as the drift speed over the Fermi velocity times the size of the flow over the electron-electron mean free path, and values of order ten have been judged the most that can be expected [54, 55]. What is added here is what a Tesla valve delivers inside it. Third, the low viscosity needed for a large Reynolds number comes with weak slip relative to the width and strong relative damping. Figure 3(b) shows the valve scaled to 1 μm: each factor of four in Da costs a factor two to three, and a slip length of 0.4 μm instead of perfect slip a factor twenty. Inside the admissible window the diodicity is 10^{-3} to 3×10^{-2}, at a resistance of 0.3 to 1.6 kΩ; the 175 nm valve costs 10 to 86 kΩ for less.

These statements are for graphene, for steady laminar flow and below the drift limit, and an experiment in another material shows what lies outside them. In a GaAs electron gas at 4 K, with channels of 2.5 and 5 μm, the forward and reverse characteristics of a Tesla valve coincide up to about 300 μA and then separate, the reverse resistance rising to a diodicity of about 40; the authors estimate a Reynolds number of at least 7 and attribute the effect to the onset of turbulence [19]. Below the threshold this is the reciprocity found here at low Reynolds number. Above it, the steady solution of Eq. (3) gives $D - 1$ of order 0.1 at $\mathrm{Re} = 7$ to 10 with slip walls and less without, two orders of magnitude short, so the large diodicity is not the steady inertial loss computed in this section. Unsteady flow, the heating of the electrons that the authors report and a drift speed that is a sizeable fraction of the Fermi velocity are all outside Eq. (3), and we make no statement about that regime.

C. With the gate: a circuit term and no shape term

Equation (3) has no gate. In a gated device the density follows the local potential, so a bias opens the channel at one end and closes it at the other, and a second nonlinearity appears that owes nothing to inertia. We computed the steady states of Eqs. (5) and (6) for both funnels, both Tesla valves, a straight channel and a mirror-symmetric constriction, with a current source at one contact and the other contact tied to the gate, for the two parameter sets of Table I and biases from 1.4 to 430 mV (10^{-3} to 0.3 of the gate overdrive $U_0 = en_0/C_g = 1.43$ V).

A two-terminal device with a gate can be reversed in two ways, and they measure different things. Reversing

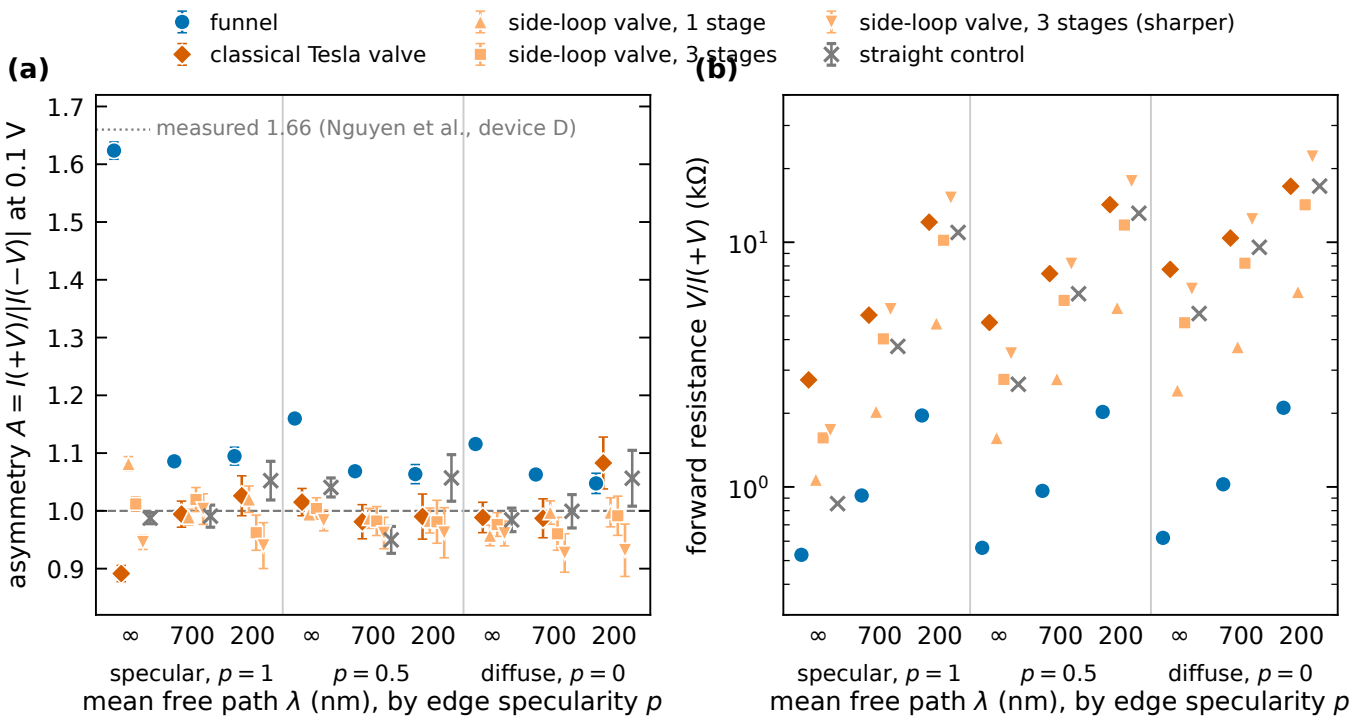


FIG. 2. Ballistic transport at dc. (a) Asymmetry $A = I(+V)/|I(-V)|$ at 0.1 V for the funnel, four Tesla-valve geometries and the straight control, for edge specularities p and mean free paths λ; error bars are one standard error from 64 batches of 400 000 trajectories in total. The dotted line marks the measured value for the funnel [4]. (b) Forward resistance of the same devices.

the polarity of the source, with the gate connection where it was, changes the sign of the bias relative to the gate. Exchanging the contacts, so that the gate is tied to the other end and the same current is sent through the shape in the other direction, leaves the bias relative to the gate as it was and reverses only the flow through the geometry. The first operation tests the circuit, the second the shape. In a gradual channel of width $W(x)$, without inertia and viscosity, the steady equations integrate to (Appendix C)

$$G(n_s) - G(1) = \Gamma I \int_0^L \frac{dx}{W(x)}, \qquad G'(n) = n\,h'(n), \quad (9)$$

in the units of Appendix A, with I the current and n_s the density at the driven contact. The shape enters through the integral, which is the linear resistance, and through nothing else: exchanging the contacts gives no asymmetry at all, and reversing the polarity gives the same asymmetry for every shape,

$$D_{\mathrm{pol}} - 1 = \frac{V}{2V_u} = 0.48\,\frac{V}{U_0} \qquad (10)$$

to first order in the bias V for the Dirac closure (V/U_0 for shallow water), the harder polarity being the one that depletes the channel.

The two-dimensional solutions bear this out [Fig. 4(a)]. Under reversal of the polarity all six devices lie on one curve, 1.45% at 43 mV, within 2% of the closed form wherever the drift speed is below $0.3\,v_F$ (4% for the 1 μm Tesla valve at its largest bias, where inertia enters). Under exchange of the contacts the asymmetry is at most 4×10^{-4} at any computed point inside the drift limit and, growing there as the square of the current, stays below 10^{-3} up to the limit; where it is above the discretisation floor it is the inertial term of the previous subsection, unchanged by the gate [Fig. 4(b)]: at 21 of the 23 points at which the incompressible diodicity exceeds 10^{-4} the gated solution agrees with it within a factor two at the same Reynolds number, damping, wall condition and mesh, and the two exceptions are one point outside the drift limit and one at the floor of its mesh. At low bias the exchange asymmetry on the production meshes is linear in the bias, which Eq. (9) forbids. It is numerical: for the funnel its coefficient falls by a factor of forty-five as the wall mesh is refined from 28 to 10 nm, while the quadratic, inertial coefficient does not move [Fig. 4(c)]. We therefore report the low-bias shape term of a gated channel as a bound, 7×10^{-6} at 14 mV for the funnel and 1.5×10^{-7} for the Tesla valve.

Three things follow. A gated channel of any shape, a straight strip included, shows an asymmetry of a per cent at some tens of millivolts when its polarity is reversed, and a measurement or a simulation that reverses the polarity will find it in a Tesla valve without the valve having anything to do with it. The compressible simulation of Ref. [18], which gives a diodicity of 1.07 for a graphene Tesla valve with a 10 nm channel at 50 mV, is not reproduced here as an effect of the shape: at 50 mV the

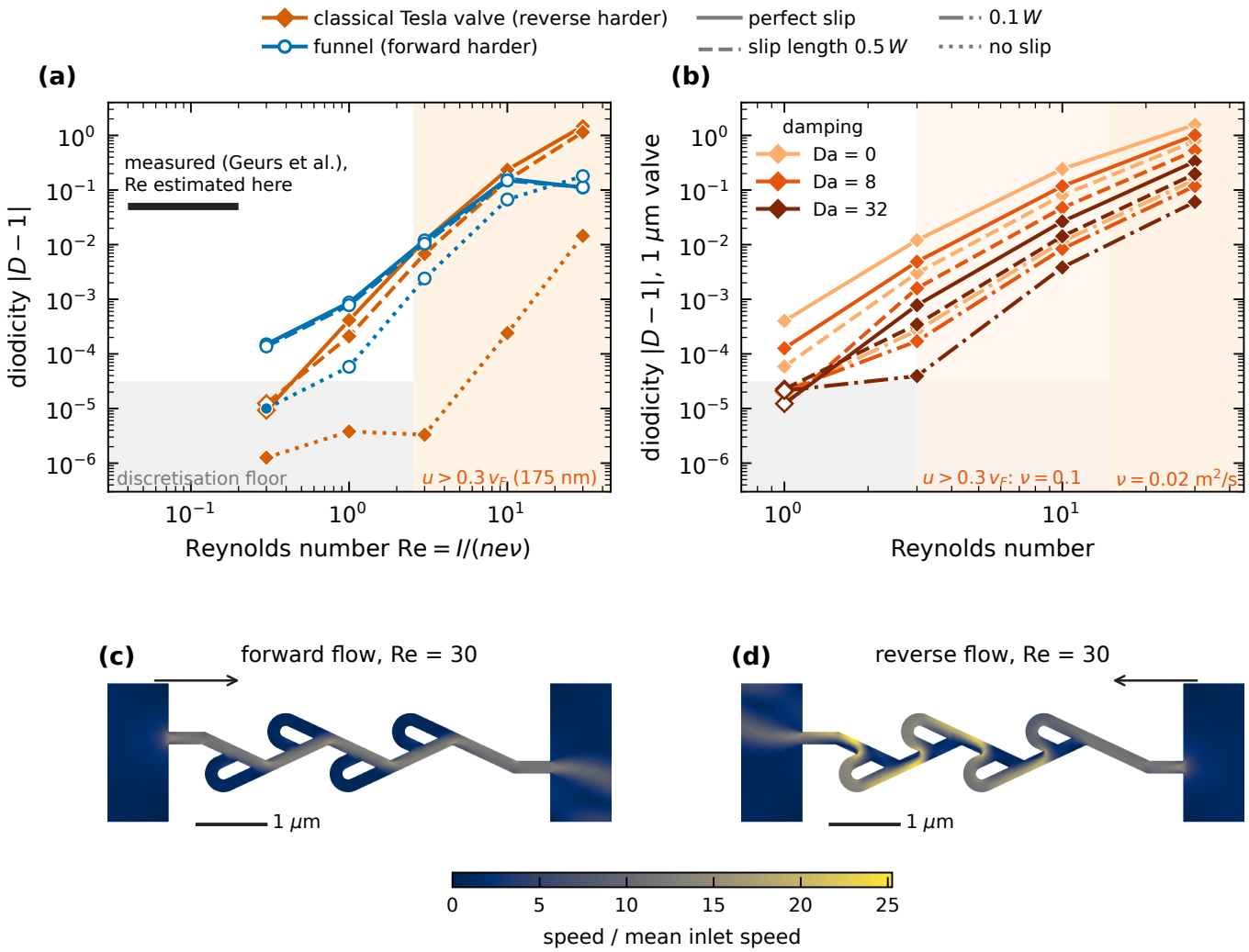


FIG. 3. Steady viscous flow. (a) Diodicity $|D-1|$ of the 175 nm classical Tesla valve and of the funnel without damping, for three wall conditions; filled symbols: reverse direction harder ($D>1$), open symbols: forward harder. The bar marks the diodicity reported in Ref. [17] at our estimate of its Reynolds number; grey: below the discretisation floor; orange: drift speed above $0.3\,v_F$. (b) The valve scaled to a 1 μm channel, with momentum-relaxation damping Da and finite slip length; the two shaded ranges are the drift-speed limits for $\nu = 0.1$ and 0.02 m^2/s. (c), (d) Speed in the 175 nm valve at Re = 30 with perfect slip for forward and reverse flow.

exchange asymmetry of our valve is below 10^{-5}, and the polarity term of Eq. (10) is 1.7% for any channel under our gate. Second, the circuit term is not offered as an explanation of the measurement of Ref. [17], which is a ratio of four-terminal resistances and is reported to vanish outside the viscous regime; that observation remains unexplained by the gated model as well. Third, the result depends on the monolayer: the plasma speed under a 30 nm gate is $3.6\,v_F$, so the drift limit keeps the flow below a tenth of the wave speed. In bilayer graphene, where the flow can be driven to the wave speed, a nozzle does show a polarity-dependent resistance of hydrodynamic origin [56], which is outside the present model.

The steady-flow asymmetries are also slow. The inertial one is established over the viscous and convective times of the channel, 10 to 20 ps at 1 μm, so the picture of this section applies below some tens of gigahertz. Above that the response of a gated channel is carried by plasma waves, and the question has to be asked of the time-dependent form of the gated model.

IV. THE PLASMONIC ELEMENT AT TERAHERTZ FREQUENCIES

A. Small signal: the terminations rectify, the shape tunes

The solver is first checked where an analytic answer exists. For a uniform channel driven by an ideal source with an open drain, the shallow-water closure reproduces the closed form of Dyakonov and Shur [26] to three dig-

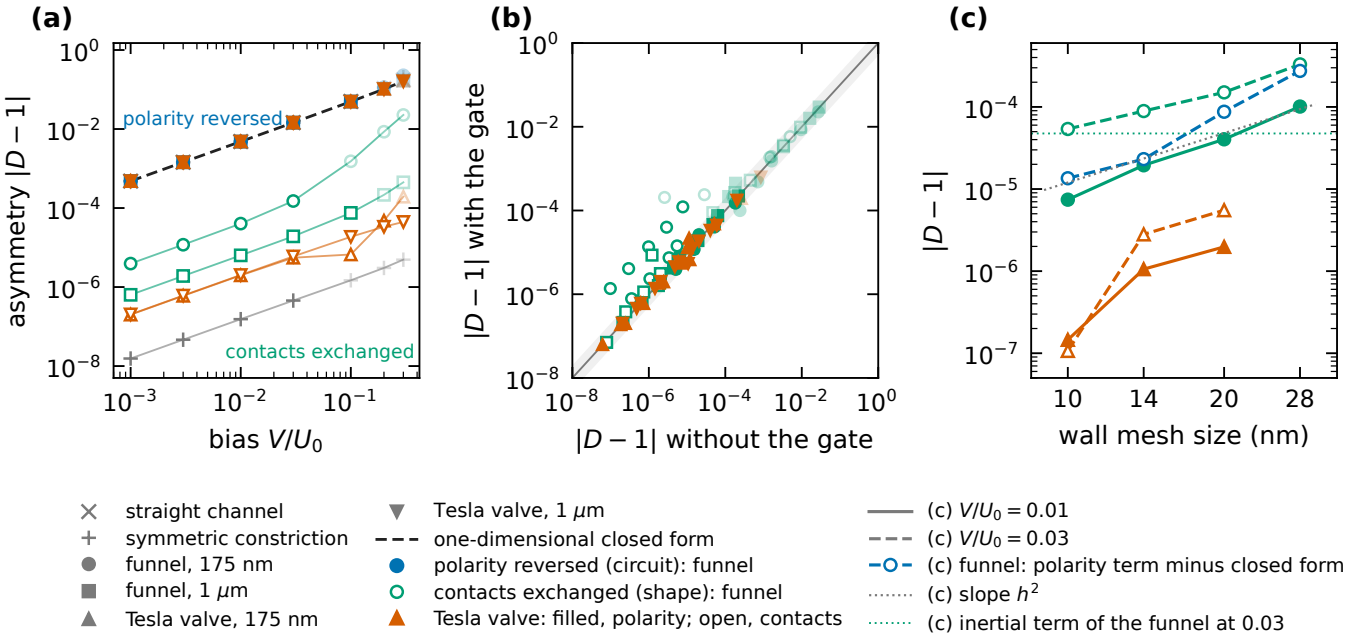


FIG. 4. Steady flow with the gate, $n_0 = 10^{12}\,\mathrm{cm}^{-2}$ under 30 nm of hBN. (a) Asymmetry of the voltage against bias for the 300 K parameters and slip walls, under reversal of the polarity (filled symbols) and under exchange of the contacts (open symbols and crosses); the dashed line follows from Eq. (9). The symmetric constriction under exchange of the contacts measures the asymmetry of its unstructured mesh. (b) Exchange asymmetry with the gate against the diodicity of the incompressible model of Eq. (3) at the same Reynolds number, damping, wall condition and mesh, for both parameter sets; filled: slip walls, open: no slip; the band is a factor two. (c) Refinement of the wall mesh for the 175 nm funnel and Tesla valve at 300 K. Faint symbols in (a) and (b): drift speed above $0.3\,v_F$.

its from 10 GHz to 3 THz, through the plasmon resonances at 100 K and on the broadband plateau at 300 K [Fig. 5(a)]. The Dirac closure of Eq. (7) raises the response by 40% near the resonances and by up to a factor two between them; the difference is the nonlinearity of the Dirac fluid itself, whose density-dependent mass and quantum capacitance add second-order terms that the shallow-water model lacks.

Figure 5(b) shows the 1 μm funnel. Its impedance is 80 to 200 Ω across the band, so it absorbs 80 to 95% of the available power of a 75 Ω source at 100 K. With the DS termination the responsivity is 0.15 to 0.34 A/W at 100 K and 0.07 to 0.09 A/W at 300 K, flat from 10 GHz to beyond 1 THz. With the symmetric termination, which isolates the shape, it is 10^{-5} to 10^{-2} A/W below 100 GHz, rises to 0.13 A/W at 1 THz at 100 K as the taper becomes comparable to the plasmon wavelength, and stays at a few mA/W at 300 K, where friction and viscosity damp the plasmon. Below 100 GHz the shape-only response is the curvature of the steady current-voltage relation of Sec. III; a ladder test ties the two solvers together there. The sign of the DS response follows the termination: exchanging the roles of source and drain reverses it. That the shape of a channel modifies the response set by its boundary conditions, rather than replacing it, is in line with what is known for silicon and graphene field-effect detectors: a non-uniform channel changes the responsivity by about ten per cent [57], and a Corbino geometry, which concentrates the field at the inner contact, raises it substantially [58].

The 175 nm funnel [Fig. 5(c)] has a dc resistance of 1.9 kΩ and absorbs only 15 to 80% of the available power, but it has one feature the wide device lacks. At 1.87 THz the taper passes through a series resonance, the horn is matched to the antenna, and the shape-only responsivity reaches 0.22 A/W at 100 K, half of the 0.45 A/W of the DS termination at the same frequency. The resonance belongs to the taper and not to the leads: a control with 1 μm leads has it at the same frequency, with an impedance of $57 - 8i$ Ω there. This is the one place where the funnel shape earns its keep in a plasmonic design, as a matched horn. At 300 K it gives 0.03 A/W against 0.10 A/W for the terminations.

The Tesla valve [Fig. 5(d)] contributes almost nothing. Its shape-only responsivity is 10^{-5} to 10^{-3} A/W below 300 GHz and reaches 2×10^{-2} A/W only at a 100 K resonance near 0.7 THz, ten to twenty times below the funnel at every frequency and temperature. Its resistance of 6 to 11 kΩ is two orders of magnitude from the antenna, so 3 to 50% of the available power is absorbed, and whatever the DS termination adds, 0.02 to 0.1 A/W, is the terminations and not the loops. The valve's asymmetry is an inertial loss that needs a Reynolds number of ten; at terahertz frequencies the long path that produces it is also several plasmon wavelengths long, so its stages do not act coherently.

With ideal contacts and the Johnson noise of $R_{\rm dc}$ alone, the noise-equivalent power of the funnel elements is 4 to 60 $\mathrm{pW/Hz^{1/2}}$ between 0.3 and 2 THz: 17 $\mathrm{pW/Hz^{1/2}}$ for the 1 μm funnel at 100 K, about 60 $\mathrm{pW/Hz^{1/2}}$ at 300 K, and 4 $\mathrm{pW/Hz^{1/2}}$ for the 175 nm funnel at its taper resonance at 100 K.

B. Large signal: a square law up to the gate overdrive

A detector is judged by responsivity and noise, a harvester by conversion efficiency $\eta = P_{\rm dc}/P_{\rm av}$ at the power the antenna supplies. The framework is that of Kachorovskii *et al.* [32] for field-effect detectors, who define the same ratio with respect to the absorbed power, find it quadratic in the amplitude at small signal, and show that it has an absolute maximum of the order of ten per cent, reached below threshold where the voltage scale is the thermal one; the change from quadratic to linear response at an amplitude equal to the gate swing is established in theory and experiment [29–31]. What is specific to graphene is that a gapless monolayer has no subthreshold regime: the only voltage scale of the gated fluid is the gate overdrive, and the question is how large the pure-number prefactor is in a real geometry with a real antenna. In second-order perturbation the device is a dc Thevenin source with open-circuit voltage $V_{\rm oc} = r\,V_a^2/V_u$ and resistance $R_{\rm dc}$, where $V_u = m^* s_0^2/e$ is the natural voltage unit of the gated fluid (1.49 V here, the gate overdrive up to the quantum-capacitance correction) and r is a pure number. Into a matched dc load

$$\eta = 2r^2\,\frac{R_{\rm ant}}{R_{\rm dc}}\,\frac{V_a^2}{V_u^2} = \frac{16\,r^2 R_{\rm ant}^2}{R_{\rm dc}V_u^2}\,P_{\rm av}. \qquad (11)$$

For the 1 μm funnel $r \approx 0.1$ and $\eta/P_{\rm av}$ is 1.0 to 2.4 $\mathrm{W^{-1}}$. Since V_a cannot usefully exceed V_u, where the channel is depleted on the negative half cycle, Eq. (11) already bounds the efficiency below one per cent; what the nonlinear solver decides is where the law fails and how. The prefactor $2r^2R_{\rm ant}/R_{\rm dc}$ is 0.004 to 0.009 at 0.1 and 0.3 THz.

Figure 6 shows the time-domain results for the DS-terminated 1 μm funnel. The efficiency meets the perturbative law to 0.3% at the smallest amplitude and stays within about 10% of it up to 0.9 mW of available power at 0.1 and 0.3 THz at both temperatures. At 1 THz the response saturates first, 10% below the law at 150 μW at 100 K. At 1.8 mW, a swing of 1.04 V on a gate overdrive of 1.43 V, the efficiency is 0.14 to 0.38% and the response is 0.6 to 0.9 of the square law at 100 K; at 300 K the overdamped device is weakly super-quadratic, 1.10 to 1.14 of the law at 0.1 and 0.3 THz. The density inside the device swings between 0.52 and 1.6 of its gated value [Fig. 6(c)] and the solver converges at every step, so the saturation is a property of the fluid and not a failure of the method. Returning the dc current through the antenna instead of a matched load costs the expected 15% and nothing more.

The shape-only response behaves worse, not better, at large signal. With the symmetric drive the funnel gives 0.56 $\mathrm{W^{-1}}$ at 0.3 THz, a third of the DS term of the same device; it leaves its square law at 150 μW and is at 0.43 of it at 0.9 mW [Fig. 6(b)]. The picture of a geometric diode that improves when driven hard is not what the fluid does: the constriction's rectification is a resonance effect that detunes as the density swings.

The scale of the result is set by Eq. (11). A beam of 1 $\mathrm{W/m^2}$ delivers about 10 nW to a half-wave antenna at 1 THz, which the element converts at $\eta \sim 10^{-8}$; a full milliwatt focused on one element is converted at 0.2%.

C. Operating point and contacts: the knee moves, the ceiling does not

Equation (11) suggests a lever. The efficiency saturates when V_a approaches V_u, at an available power $P_{\rm knee} = (V_u/2)^2/8R_{\rm ant}$, and V_u is proportional to the gate overdrive $en_0/C_g \propto n_0 d$. A lower density or a thinner gate should bring the saturation down to the microwatts an antenna collects. What it does to the peak depends on the pure number $r^2R_{\rm ant}/R_{\rm dc}$, which the solver has to supply. We swept $n_0 = 5\times10^{10}$ to $10^{12}\,\mathrm{cm^{-2}}$ and $d = 30$, 10, 5 nm with the scattering times of Table I, flagging points outside the Fermi-liquid limit $E_F \geq 3k_BT$ or the gating limit $en_0/C_g \geq 4k_BT/e$; the saturation factors of Sec. IV B were carried over in units of V_u (Appendix B), and a second time-domain ladder at $(5\times10^{11}\,\mathrm{cm^{-2}}, 10\,\mathrm{nm})$ confirmed that they transfer, to 0.2% at 300 K and within 14% at 100 K.

The saturation power moves as predicted [Fig. 7(a)]: from 0.92 mW at the reference point to 33 μW at $(5\times10^{11}\,\mathrm{cm^{-2}}, 10\,\mathrm{nm})$ and 3 μW at $(5\times10^{10}\,\mathrm{cm^{-2}}, 30\,\mathrm{nm})$, with $\eta/P_{\rm av}$ rising from 1.9 to 51 and 1770 $\mathrm{W^{-1}}$ at 100 K and 0.3 THz. The peak barely moves [Fig. 7(b)]. Over the valid grid it is 0.03 to 0.9% at 100 K and at most 0.24% at 300 K, where the reference point is already the best. A thinner gate lowers it, because r falls from 0.088 to 0.050 between 30 and 5 nm as the plasma speed drops and friction and quantum capacitance gain weight; a lower density raises r, to 0.32 at the edge of the Fermi-liquid limit. A matched antenna in place of 75 Ω gives at best 1.3% at 100 K and 0.27% at 300 K. If the mobility rather than the scattering time is held constant as the density falls, the low-density gain shrinks by a factor four to six. The design rule is therefore to put the knee at the available power; the element then converts at a few tenths of a per cent, and nothing in n_0 or d raises that number.

Contact resistance is the dominant practical loss, and it is not a series loss. A contact resistance R_c at each contact of the reference device reduces $\eta/P_{\rm av}$ by factors 1.6, 4.6 and 78 for $R_c = 100$, 300 and 1000 $\Omega\,\mu$m at 100 K and 0.3 THz [Fig. 7(c)], the responsivity from 0.21 A/W

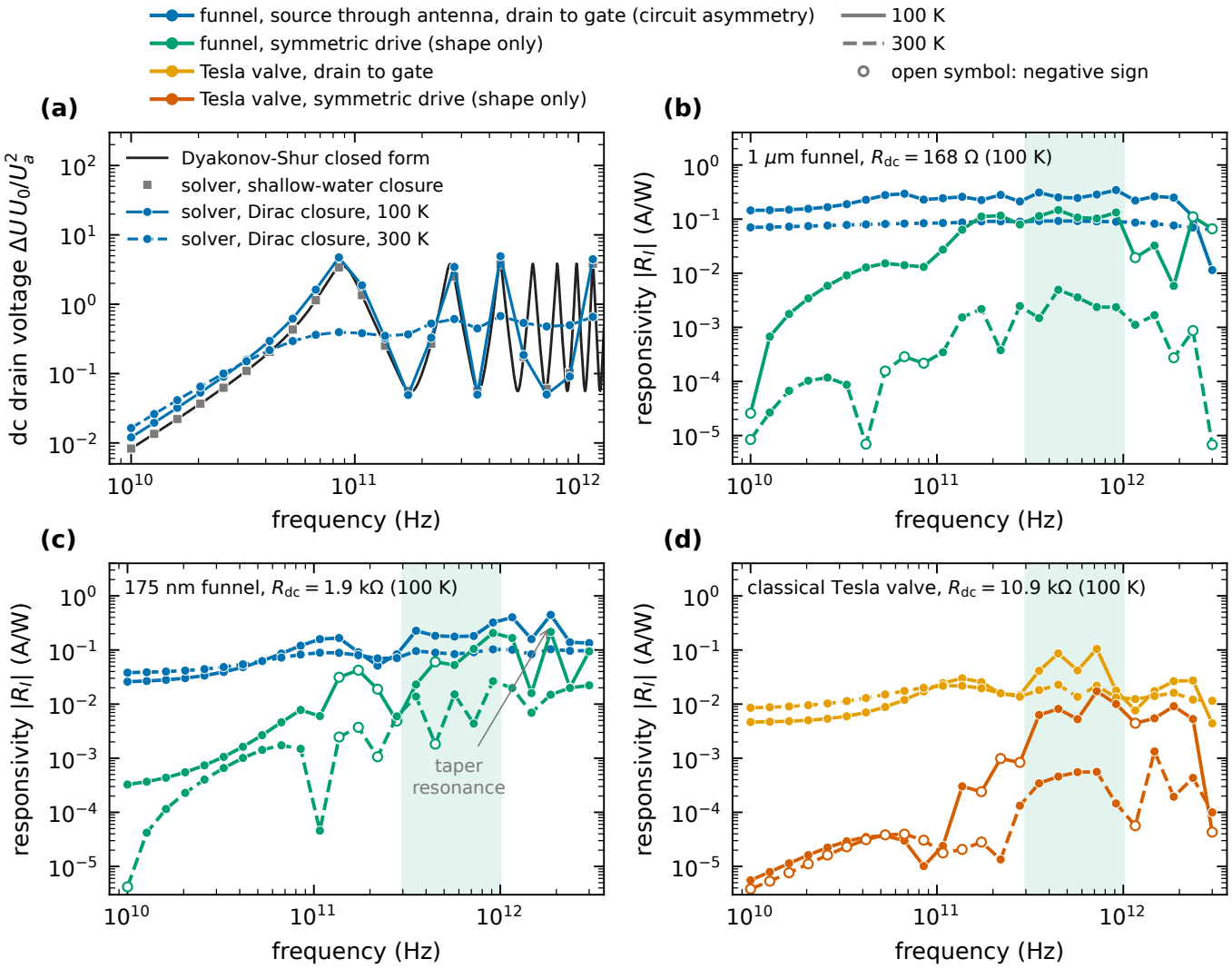


FIG. 5. Small-signal plasmonic response. (a) Validation on a uniform 1 μm × 10 μm channel with an ideal source and open drain: dc drain voltage per squared ac amplitude in units of the inverse gate overdrive. The solver with the shallow-water closure lies on the Dyakonov-Shur closed form [26]; the Dirac closure of Eq. (7) is 40% higher near resonance. (b)–(d) Current responsivity into a 75 Ω antenna for the 1 μm funnel, the 175 nm funnel and the classical Tesla valve, with the drain tied to the gate (circuit asymmetry) and with the symmetric drive that isolates the shape. Open symbols mark a negative sign. The shaded band is 0.3 to 1 THz.

to 0.15, 0.07 and 0.01 A/W, and it raises the NEP from 27 to 35, 59 and 240 pW/Hz$^{1/2}$. The series picture underestimates this. The gate carries displacement current, so the ac drain current is not the negative of the source current, and a resistance at the drain lifts it off the gate potential at the signal frequency: the boundary condition that makes the Dyakonov-Shur mechanism work is partly removed. Matching the antenna to the new impedance recovers little. That a lower access resistance raises the photoresponse of graphene detectors strongly, by up to a factor twenty for a tenfold reduction, has been measured and described with a series-resistance model [59], and that the shunt between gate and drain at the signal frequency is what makes a transistor mix is the basis of the distributed self-mixing picture [60]; the present result connects the two.

For detection the same lever is worth having [Fig. 7(d)]. The Johnson-limited NEP of the 1 μm funnel at 0.3 THz falls from 27 pW/Hz$^{1/2}$ at the reference point to about 1 pW/Hz$^{1/2}$ at 5×10^{10} cm^{-2} at 100 K if the 5 ps scattering time survives there, 2 pW/Hz$^{1/2}$ with constant mobility or with 300 Ω μm contacts; at 300 K the best valid point is 15 pW/Hz$^{1/2}$.

V. THE BALLISTIC ELEMENT AT TERAHERTZ FREQUENCIES

One mechanism is outside the fluid model by construction. In the clean limit the ballistic funnel of Sec. III has

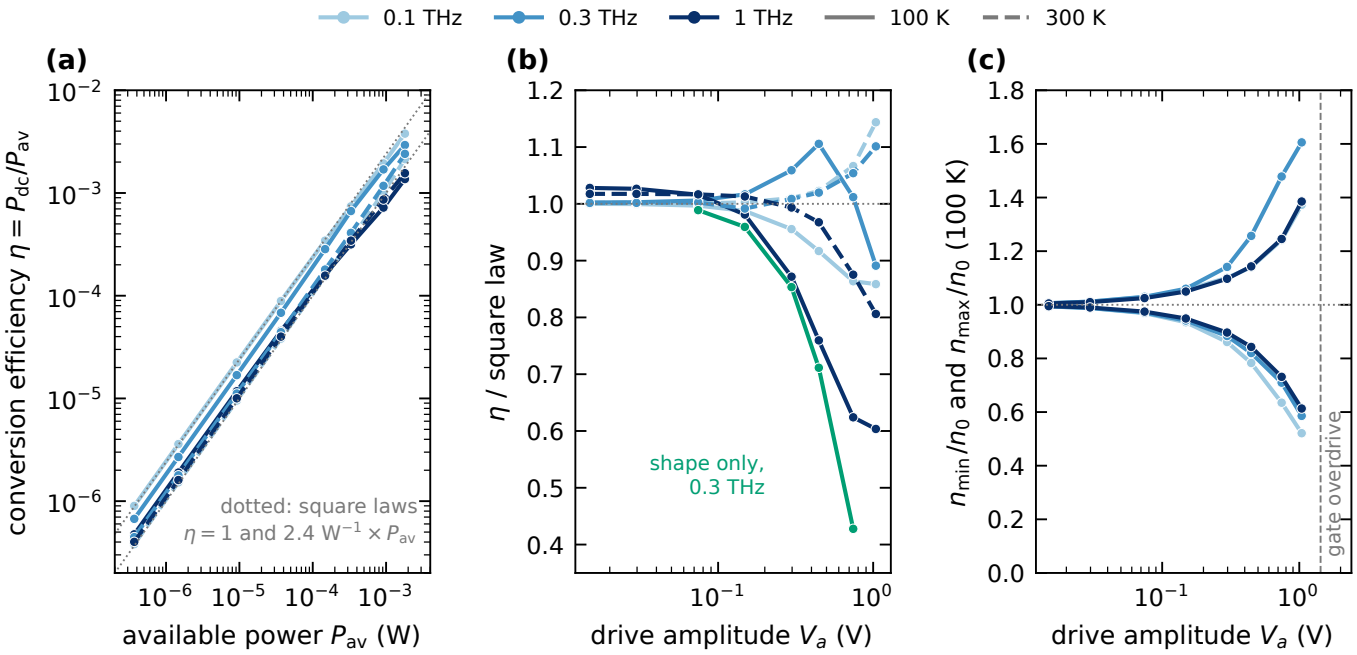


FIG. 6. Finite-amplitude response of the DS-terminated 1 μm funnel from the time-domain solver. (a) Conversion efficiency into a matched dc load against available power. (b) Efficiency relative to the square law of Eq. (11), against the ac amplitude; the green curve is the shape-only response (symmetric drive) at 0.3 THz, 100 K. (c) Smallest and largest density in the device over a period, relative to the gated density, at 100 K.

a curvature $G_2 = 3.8$ mA/V^2; if it responded instantaneously, a cosine drive of 0.1 V across the device would rectify 18 μA, which is 0.87 A/W of the available power of a 75 Ω source, two to ten times the plasmonic figures. Whether it does respond is a transit-time question. It has been asked before of a diode of about 100 nm, with a time-dependent particle simulation that found rectification into the terahertz range and none above about 5 THz [11]; we ask it of the device with the documented dc asymmetry, then of its length, and for the shortest device we compute the field as well.

The imposed potential is made time dependent, $U(x,t) = -eV_a \cos(\omega t)\, g(x)$, and the flights integrate $\hbar\dot{\boldsymbol{k}} = -\nabla U$ without the energy conservation that holds only in a static landscape. Two profiles g bracket the electrostatics, which the model does not compute: the uniform ramp over the 6 μm device used at dc, and a drop of length ℓ_F centred on the neck, with $\ell_F = 175$, 500 and 1600 nm (one neck width, three, and the taper length). The bias-window estimator of Eq. (1) relies on flux conservation in a fixed landscape and fails in a moving one, so electrons are injected from both contacts with the equilibrium distribution $N(E)f(E-E_F)$ at uniformly distributed phases, and

$$I_{\rm dc} = \frac{ge}{h}\,\Phi\,\big[\langle T_{SD}\rangle - \langle T_{DS}\rangle\big], \tag{12}$$

with $\Phi = \int dE\, N(E) f(E-E_F)$ and the averages taken over injection phase; $(ge/h)\Phi$ is 1.43 mA for the 2 μm contacts. The first harmonic of the drain current, from the arrival phases, gives the transport admittance and hence the mismatch to the antenna. Six further tests validate the estimator: it reproduces the bias-window current in a static field, it gives zero at zero bias, zero for the straight control at every frequency, and at 300 MHz it reproduces the rectified current and the admittance computed from the static curve (Supplemental Material). The results are reported as the transfer $S_B(f) = I_{\rm dc}(f)/I_{\rm dc}(0)$, with $I_{\rm dc}(0)$ the quasi-static value for the same field profile, and as responsivity in two forms. R_I is the current of one device per available power of the 75 Ω source, with the mismatch taken from the first-harmonic admittance. The ballistic devices have resistances of 0.6 to 4 kΩ, far from the antenna, so we also give the current per watt absorbed at the static conductance G,

$$R_{\rm abs} = \frac{2\,I_{\rm dc}}{V_a^2 G}, \tag{13}$$

which is what $N = 1/(R_{\rm ant}G)$ devices in parallel would give on a matched antenna if their impedance did not change with frequency. The plasmonic funnel of Sec. IV absorbs 87 to 100% of the available power above 0.1 THz, so its R_I is to be compared with $R_{\rm abs}$. We do not normalise by the power that the model itself absorbs at the drive frequency: with an imposed field and no displacement current its conduction admittance falls, vanishes and changes sign with frequency, and a responsivity so defined diverges where it does. For the one device whose field is computed (Sec. V E) the absorbed power is a result, and $R_{\rm abs}$ there is the rectified current over it.

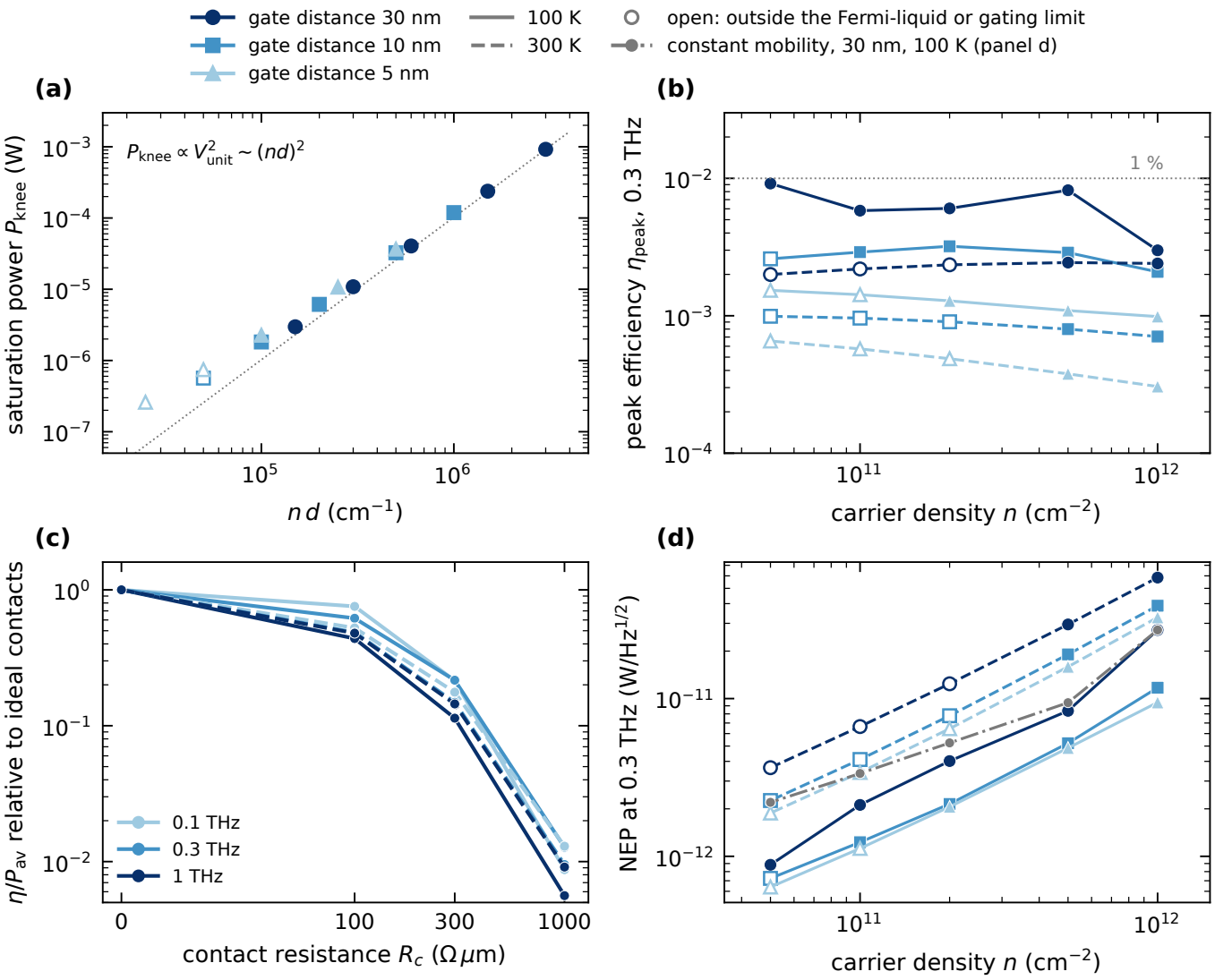


FIG. 7. Carrier density, gate distance and contacts for the DS-terminated 1 μm funnel. (a) Saturation power P_{knee} against the product $n_0 d$; the dotted line is $(n_0 d)^2$ through the reference point, and the departure at small $n_0 d$ is the quantum capacitance. (b) Predicted peak efficiency at 0.3 THz. (c) Efficiency per watt against contact resistance, relative to ideal contacts, at the reference point. (d) Johnson-limited NEP at 0.3 THz. Open symbols are outside the Fermi-liquid or gating limits.

A. The 6 μm funnel in a uniform field: two roll-offs

With the uniform field the response dies in two steps [Fig. 8(a)]. The transfer is 0.98 ± 0.04 at 0.3 GHz and already 0.82 at 3 GHz and 0.67 at 10 GHz, although a direct flight from the source contact to the neck takes 3.6 ps. The reason is in the distribution of flight times [Fig. 8(c)]. The taper scatters part of the incoming electrons into directions nearly perpendicular to the axis, and these drift back to the source contact, bouncing between the walls of the lead, over tens of picoseconds: 5% of the source-injected electrons fly longer than 20 ps and 0.9% longer than 50 ps. In a field that fills the leads, this slow population is displaced in energy between wall collisions once the period approaches its lifetime, and the transmission in both directions changes. The response time of a ballistic device is set by its slowest population, not by its mean transit, and a converging taper manufactures slow electrons. The second step is the transit roll-off of the direct electrons: the transfer is 0.41 at 100 GHz, crosses zero at 300 GHz, where $\sin(\pi f \tau_{tr})$ vanishes for $\tau_{tr} = 3.6$ ps, and is 0.05 ± 0.04 at 1 THz, i.e. 0.05 A/W. At a mean free path of 700 nm the quasi-static responsivity is only 0.15 A/W to begin with; it is 0.04 A/W at 30 GHz and zero at 1 THz. With the field model in which the dc asymmetry was established, the 6 μm funnel is not a terahertz rectifier.

B. Field confined to the neck

Self-consistent electrostatics concentrates the field of such a diode at the neck rather than spreading it along the leads [35]. Confining the drop to the neck lowers the static asymmetry, from 1.61 for the uniform field to 1.52, 1.50 and 1.40 for $\ell_F = 1600$, 500 and 175 nm, and it changes the dynamics [Fig. 8(a) and 8(b)]. For $\ell_F = 500$ nm the transfer is 1.02 ± 0.04 up to 3 GHz, since the slow electrons in the lead no longer see a field, then falls to 0.54 at 100 GHz, and is 0.64 at 300 GHz and 0.38 ± 0.04 at 1 THz. The decline between 10 and 100 GHz is not what the 0.5 ps crossing of the field region would suggest; the transfer is set by the whole flight between the contacts. At 1 THz the transfer is 0.51 for a drop of one neck width, 0.38 for 500 nm and zero for 1600 nm. The corresponding responsivities of one device are 0.45 ± 0.04 and 0.37 ± 0.04 A/W, as much as the Dyakonov-Shur response of the plasmonic funnel at 100 K; per absorbed watt the 500 nm drop gives 0.72 ± 0.07 A/W, about twice that. The amplitude dependence is that of the dc curve: at 300 GHz the rectified current follows V_a^2 to 0.1 V and is 0.40 and 0.24 of the square law at 0.2 and 0.3 V.

Two further cases show that the terahertz response of the confined field is not simply the static asymmetry carried up in frequency. At $\lambda = 700$ nm the static asymmetry of the confined-field models is reversed ($A = 0.94$ to 0.97), the rectified current passes through zero between 30 and 100 GHz, and it returns with the sign of the clean funnel, 0.20 A/W at 300 GHz and 0.24 A/W at 1 THz (0.66 and 0.75 A/W per absorbed watt). With half-diffuse walls ($p = 0.5$) in the clean limit the static asymmetry of the 500 nm drop is 1.01, and the responsivity at 1 THz is nevertheless 0.45 ± 0.04 A/W (0.31 A/W for the neck-width drop). There is, then, a transit-time rectification of the funnel with a neck-confined field that is robust against edge roughness and bulk scattering, of 0.2 to 0.45 A/W for one device. It does not depend on the treatment of the few electrons that reach the Dirac point: replacing their removal by a classical turning point changes no result by more than one standard error. It does depend on the confinement of the field, which this model assumes and cannot test: at 1 THz and $\lambda = 700$ nm the 6 μm device gives 0.00 ± 0.04 A/W per absorbed watt with the uniform field and 0.75 ± 0.05 A/W with the 500 nm drop. For this device the terahertz response is decided by electrostatics that the model does not contain.

C. Length scaling

The two roll-offs of the 6 μm device are properties of its size. The model has no length of its own (Sec. II B): a copy of the device at half scale has half the current at every bias, and under a drive at twice the frequency it has half the rectified current. Both statements were tested, as was the equivalence with scattering at fixed λ/L: the static current of device D at 0.1 V and twice that of its half-scale copy are 188.0 and 188.6 μA, and the rectified current at 10 GHz and twice that of the copy at 20 GHz are 11.9 ± 0.4 and 12.1 ± 0.5 μA. The frequency response of a given shape is therefore a function of the reduced frequency fL/v_F, and the question whether a ballistic funnel rectifies at 1 THz becomes the question how long it is.

Table II and Fig. 9(a) define five devices: the 6 μm funnel (G0), its half-scale copy (G1), the same taper and neck with the leads cut to 250 nm (G2), a 0.55 μm funnel with a 100 nm neck (G3), and a 0.50 μm funnel with the shoulder and neck of the diode measured at 28 THz in Ref. [2], for which we assume a 45 degree taper (G4). Mirror-symmetric versions of G2 to G4 serve as controls; at 0.1, 1 and 10 THz and for both field profiles all 22 control points are zero within 2.1 standard errors. The confined drop is one neck width long in every device.

With a uniform field and no scattering the five transfer functions fall on one curve [Fig. 9(b)]: above $fL/v_F = 0.2$ they agree within about 0.2, the response is half its quasi-static value near $fL/v_F = 0.4$, and it is zero from about 1.7 on; for device D it remains zero within 0.04 out to $fL/v_F = 30$. The frequency of half response is thus $0.4\, v_F/L$: 60 GHz for the 6 μm device and 0.7 THz for the 0.55 μm one, and the response is gone at 280 GHz and at 3 THz. The same scaling with the size of the scatterer over the Fermi velocity was found for the ratchet current of Dirac electrons on asymmetric antidots [61]. Below $fL/v_F = 0.2$ the curves differ, and the difference identifies the gigahertz step of the 6 μm device. Devices G0 and G1, with leads as long as the taper, share a plateau at 0.67 between $fL/v_F = 0.03$ and 0.3; G2 and G3, with short leads, are still at 0.94 and 0.90 at 0.06, and G4 lies between. The slow population belongs to the leads, as inferred above from the flight times. Scattering lowers the cut-off when the device is longer than the mean free path and not otherwise: for G3 at $\lambda = 700$ nm ($\lambda/L = 1.3$) the transfer is that of the clean device, 0.76 and 0.32 at 0.3 and 1 THz, while for G2 ($\lambda/L = 0.34$) it is 0.41 against 0.75 at $fL/v_F = 0.2$. A cut-off near the Drude damping rate, as reported for the simulated diode of Ref. [11], and a cut-off at the flight time are the two limits of the same behaviour.

D. A short funnel at 1 THz with an imposed field

For the 0.55 μm funnel the reduced frequency at 1 THz is 0.55, and the device is shorter than the room-temperature mean free path. At $\lambda = 700$ nm and 1 THz the imposed-field model gives

$$R_{\rm abs} = \begin{cases} 0.70 \pm 0.02\,\text{A/W} & \text{uniform field,} \\ 0.71 \pm 0.03\,\text{A/W} & \text{drop at the neck,} \end{cases} \tag{14}$$

and $R_I = 0.17$ and 0.14 A/W for one device of 1.2 kΩ on the 75 Ω antenna [Fig. 9(c)]; G4 gives 0.57 ± 0.03 and

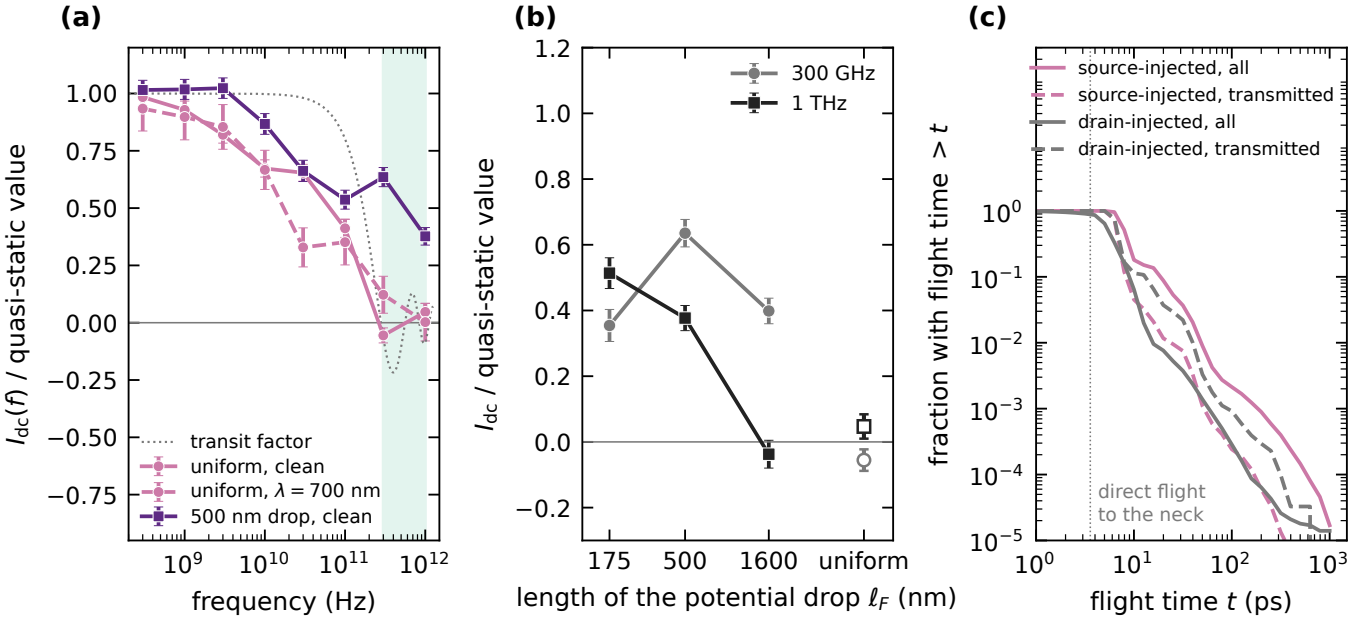


FIG. 8. The 6 μm ballistic funnel under an ac drive of amplitude 0.1 V. (a) Rectified current relative to its quasi-static value for the uniform field (clean limit and $\lambda = 700$ nm) and for a potential drop of 500 nm at the neck; the dotted line is the transit factor for a flight of 3.6 ps from the source contact to the neck. (b) The same ratio at 0.3 and 1 THz against the length of the drop; open symbols: uniform field. (c) Fraction of electrons with a flight time longer than t in the static device at 0.1 V. Error bars are one standard error.

TABLE II. Funnels used for the length scaling. Shoulder width, neck width and lead lengths (source, drain) in nm; L is the length between the contacts. A is the static asymmetry at 0.1 V with a uniform field for no scattering and for mean free paths of 700 and 100 nm; R is the resistance at $\lambda = 700$ nm and $N = 1/(R_{\rm ant}G)$ the number of devices in parallel that matches 75 Ω. G4 has the shoulder and neck of the diode of Ref. [2] with an assumed 45 degree taper; the others have the 30 degree taper of device D.

	Description	Shoulder/neck	Leads	L (μm)	A, clean	A, 700 nm	A, 100 nm	R (kΩ)	N
G0	device D of Ref. [4]	2000/175	2000, 2420	6.00	1.61	1.10		0.95	13
G1	G0 at half scale	1000/88	1000, 1210	3.00	1.61	1.13	1.06	1.50	20
G2	G0 with short leads	2000/175	250, 250	2.08	1.91	1.55	1.33	0.77	10
G3	short funnel	500/100	100, 100	0.55	1.83	1.56	1.28	1.16	15
G4	proportions of Ref. [2]	400/75	169, 169	0.50	1.60	1.30	1.15	1.54	21

0.70 ± 0.02 A/W per absorbed watt. Two things have changed with respect to the 6 μm device. The two profiles, which differed by the whole effect at 6 μm, agree within the statistical error [Fig. 9(d)]. And the result holds at the mean free path of encapsulated graphene at room temperature, not only in the clean limit, in which the same device gives 0.95 ± 0.05 and 0.83 ± 0.06 A/W. Taken at face value this is two to three times the response of the plasmonic funnel at 100 K and eight times its 0.09 A/W at 300 K.

In the same model the edges and scattering cost a factor of up to five. With half-diffuse walls ($p = 0.5$) the figures at 1 THz and $\lambda = 700$ nm are 0.42 A/W (uniform) and 0.14 A/W (confined); with a mean free path of 100 nm, which is also a rough bound for the effect of electron-electron collisions at 300 K, they are 0.22 ± 0.04 and 0.42 ± 0.04 A/W. With the confined drop the static asymmetry of G3 falls below one when scattering is strong ($A = 0.81$ at 100 nm, 0.90 with half-diffuse walls at 700 nm), while at 1 THz the current has the sign of the clean funnel in every case, as for the 6 μm device. The measured devices bear the dependence on the mean free path out in order of magnitude: the diode of Ref. [2], with a measured mean free path of about 45 nm, had a zero-bias dc responsivity of 0.12 A/W, against 0.71 A/W for G4 at 100 nm and 1.33 A/W at 700 nm in the quasi-static limit of the uniform-field model, and the rectifiers of chemical vapor deposited graphene of Ref. [9], with mean free paths of at most 65 nm, have a responsivity of 0.8 V/W referred to the beam power, against hundreds of V/W for encapsulated ballistic rectifiers [7]. The amplitude dependence is a square law with a low ceiling: with the uniform field the rectified current of G3 follows V_a^2 up to 0.1 V and is 0.80 and 0.57 of the law at 0.2

and 0.3 V, and with the confined drop it is already 19% below the law at 0.1 V. The scale is the Fermi energy, $E_F/e = 0.14$ V, a tenth of the gate overdrive that limits the plasmonic element.

Above 1 THz the two field profiles separate again. With the uniform field the response of G3 and G4 is zero at 3 THz, as the scaling requires. With the drop confined to the neck it is 0.85 ± 0.03 and 1.03 ± 0.03 A/W at 3 THz and has fallen by an order of magnitude at 10 THz, to 0.05 ± 0.03 and 0.09 ± 0.03 A/W at $\lambda = 700$ nm (0.23 ± 0.04 A/W for G4 at 100 nm); at 30 THz no device, profile or mean free path gives a current distinguishable from zero. The response at 28 THz measured for a diode of the size of G4 [2] is not reproduced. At that frequency the photon energy, 116 meV, is close to the Fermi energy and a classical trajectory in a classical field is not an adequate description; interband absorption, which the model lacks, is the natural candidate for what was measured.

The agreement of the two profiles in Eq. (14) suggests that for a device this short the field does not matter. The next subsection tests that by computing it, and finds that it does.

E. The short funnel with the field computed

We take the 0.55 μm funnel G3 into the model of Sec. II C, in which the potential follows from the charge of the electrons, a gate and metal contacts. The operating point is that of Eq. (14): $\lambda = 700$ nm, 1 THz, an amplitude of 0.1 V between the contacts, the drive that leaves only the shape, and the gate at 300 nm. The funnel rectifies 0.429 ± 0.012 μA of an absorbed 3.44 μW,

$$R_{\rm abs} = \frac{I_{\rm dc}}{P_{\rm abs}} = 0.124 \pm 0.003\,{\rm A/W}, \qquad (15)$$

and $R_I = 0.023$ A/W for one device on the 75 Ω antenna, with an admittance of $0.69 - 0.17i$ mS. Under the 30 nm gate the figures are 0.132 ± 0.003 and 0.026 A/W, with $0.73 + 0.34i$ mS. The absorbed power is now a computed quantity; in the definition of Eq. (13), for comparison, the values are 0.10 and 0.11 A/W against 0.70 A/W. The errors are statistical, from seven and six runs. Changes of the grid, the time step, the particle number and the length of the contacts move the current between 0.38 and 0.49 μA, and the mirror-symmetric version of the device, which must give zero under this drive, gives currents of up to 0.06 μA in single runs at 0.3 and 1 THz (the mean of six runs from 0.3 to 3 THz is 0.01 ± 0.02 μA); we take 0.05 μA as the floor of the method and quote 0.12 ± 0.015 A/W. The computed field leaves a sixth of what either imposed field gave.

The loss is already present at dc. The static asymmetry at 0.1 V is 1.053 under both gates, against 1.56 and 1.13 for the two imposed profiles, while the conductance, 0.86 mS, is that of the uniform-field model: only the even part of the current-voltage curve has changed. Figure 10(a) and 10(b) show why. About 61% of the bias drops within 90 nm of the neck, centred on it; 14% along the lead and the taper; and 25% at the two contact planes, 6 to 9% at the source and 17 to 19% at the drain, where it contributes nothing to the asymmetry. The drop at the source is of the size expected for a ballistic contact, half of the current over the Sharvin conductance of the 500 nm lead, or 8% of the bias. Neither imposed profile resembles this. The uniform ramp puts most of the bias along the taper, which is where its large asymmetry comes from, and the confined drop puts all of it at the neck: with 61% there and a rectified current that goes as the square of the bias, the confined-drop model would give 0.9 μA in the quasi-static limit, and the computed curve gives 1.07 ± 0.02 μA.

The frequency dependence is that of the uniform-field model at a seventh to a ninth of its height [Fig. 10(c)]. At 0.1 THz the rectified current is 1.04 ± 0.01 μA, the quasi-static value of the computed static curve; it is 0.94, 0.71, 0.43 and 0.15 μA at 0.3, 0.6, 1 and 1.8 THz, half at 0.9 THz or $fL/v_F = 0.5$. The transit roll-off of Sec. V C thus survives the computation of the field. The agreement of the two imposed profiles at 1 THz does not: it was the crossing of a curve that falls with frequency and one that rises, and no sign of insensitivity to the field. Above 1 THz the gate matters. Under the 300 nm gate the absorbed power falls steadily, to 0.8 μW at 3 THz, with 0.19 ± 0.02 μA rectified there and nothing at 6 THz. Under the 30 nm gate the device is a plasmonic cavity between two contacts that hold the potential, with a half-wave resonance near 3.6 THz and a full-wave one near 7 THz for the plasma speed of $3.9\,v_F$; the antisymmetric shape-only drive couples to the second, the absorbed power rises to 10.7 μW at 6 THz, and the rectified current stays at 0.35 to 0.50 μA up to 3 THz.

With a gate in the model, shape and circuit can be compared in the ballistic gas as they were in the fluid [Figs. 10(d) and 10(e)]. The circuit term is the current of a straight strip of the same length and lead width with its drain tied to the gate, the Dyakonov-Shur termination, here without collisions. Under the 300 nm gate it is 0.13, 0.34 and 0.01 μA at 0.3, 1 and 3 THz, against 0.94, 0.43 and 0.19 μA for the shape term: the shape is the larger at each frequency. Under the 30 nm gate the strip gives 0.67, 1.15 and -1.77 μA against 0.89, 0.48 and 0.50 μA. At 1 THz the circuit term is then more than twice the shape term, 0.20 A/W per absorbed watt against 0.13 A/W, and the funnel with its drain tied to the gate, which has both, gives 1.01 μA, no more than the strip. At 3 THz, near the half-wave resonance that this termination does excite, the funnel absorbs four times its low-frequency power and the strip's current is reversed; the run was repeated with a longer settling time and did not change, and we have not looked for the cause. With the gate closer than the device is long, then, the gate connection rectifies more than the shape at 1 THz in the ballistic gas as well.

How much of the shape term survives in a real device is again a matter of scattering and of the edges,

and here the computed field changes the answer of the imposed one in both directions [Fig. 10(f)]. The mean free path matters little. Per absorbed watt at 1 THz the funnel gives 0.130 ± 0.005 A/W with no scattering, 0.124 at 700 nm and 0.107 ± 0.007 A/W at 100 nm (0.145, 0.132 and 0.099 A/W under the 30 nm gate), where the uniform-field model lost a factor of three. Scattering does lower the static asymmetry, 1.09, 1.05 and 1.02, and the quasi-static current, 2.1, 1.1 and 0.23 μA, but the current at 1 THz falls much less, 0.50, 0.43 and 0.22 μA: what a long mean free path adds to the static asymmetry is not available at 1 THz, and at 100 nm the roll-off has moved above 1 THz. The conductance halves at 100 nm, so one device on the antenna does lose a factor of two, $R_I = 0.012$ A/W. The edges decide. With half-diffuse walls ($p = 0.5$, $\lambda = 700$ nm) the static asymmetry is 1.005 and the current at 1 THz is -0.07 ± 0.01 μA under the 300 nm gate and -0.13 ± 0.01 μA under the 30 nm gate, a sixth to a quarter of the specular current with the opposite sign; two independent runs agree, and the mirror-symmetric control with the same walls gives zero within the floor. The imposed fields had given 0.42 and 0.14 A/W for this case; the computed field gives at most 0.04 A/W in magnitude. The shape rectification of the short funnel requires edges that reflect specularly. Only $p = 1$ and $p = 0.5$ were computed, so how specular is not known.

The amplitude dependence differs from that of the imposed fields. At 0.05 V the current is 0.10 ± 0.02 μA under both gates, consistent with the square law through the 0.1 V point. At 0.2 V, which is above E_F/e, it is 2.56 and 2.66 μA, 1.5 and 1.4 times the square law, where both imposed fields fell below it; the absorbed power follows V_a^2 within 7%, so $R_{\rm abs}$ rises to 0.20 A/W. Into a matched dc load the conversion efficiency estimated from the short-circuit current and the static conductance, $\eta = I_{\rm dc}^2/(4GP_{\rm abs})$, is 2×10^{-5} at 0.1 V, where the device absorbs 3.4 μW, and 1.5×10^{-4} at 0.2 V and 13 μW. As a detector, $N = 1/(R_{\rm ant}\,{\rm Re}\,Y_1) = 19$ devices in parallel with their reactance tuned out absorb the available power, and the Johnson noise of their static conductance gives a noise-equivalent power of 130 pW/Hz$^{1/2}$ at 1 THz and 300 K, twice that of the plasmonic funnel at the same temperature.

The outcome for the short funnel is therefore narrower than the imposed fields suggested. A 0.55 μm funnel with specular edges rectifies at 1 THz by its shape, at 0.10 to 0.13 A/W per absorbed watt for any mean free path from 100 nm up. That is the level of the plasmonic funnel at 300 K, 0.09 A/W, and below its 0.21 to 0.34 A/W at 100 K; it is not the factor of eight of Eq. (14). With rough edges it is lost, and under a close gate the circuit term exceeds it.

VI. DISCUSSION

A. Shape and circuit, regime by regime

Table III and Fig. 11 collect the results. The Tesla valve fails in graphene in every regime, and for one reason. Its asymmetry is an inertial loss: it needs a fluid, and a Reynolds number of order ten. The ballistic electron gas is not a fluid, so the loops only add resistance. The viscous electron liquid is one, but in graphene its Reynolds number is capped by the Fermi velocity at values where the valve gives a per cent or less, and the gate adds an asymmetry that is the same for a straight strip. At terahertz frequencies the valve is long compared with the plasmon wavelength and its resistance is two orders of magnitude from an antenna. The one electronic Tesla valve with a large diodicity [19] works in another material, above a threshold current, in a regime that is not the steady laminar flow of this paper.

The funnel rectifies in every regime, and what does the rectifying differs. In the electron fluid it is the circuit. In steady flow the shape leaves an inertial asymmetry of less than 10^{-3} inside the drift limit at the parameters of Table I, the same with and without the gate, while the connection of one contact to the gate produces a per cent at 43 mV in any channel. At terahertz frequencies the gated funnel's rectification is that of its terminations: tying the drain to the gate and driving the source produces 0.1 to 0.3 A/W in a funnel, as it would in a plain channel of the same impedance, while the symmetric drive that leaves only the shape produces a comparable response only near a taper resonance at low temperature, and that response saturates first at large signal.

In the ballistic gas it is the shape, together with the potential that the bias lays over it, and the first condition for it to work at a frequency f is one of size:

$$L \lesssim 0.4\,\frac{v_F}{f}. \tag{16}$$

This is the half-response point of Fig. 9(b), for a field spread over the device; with the field computed the 0.55 μm funnel is at half response at $0.5\,v_F/L$. At 1 THz the criterion gives 0.4 μm. The 6 μm funnel on which the dc asymmetry of geometric diodes is best documented is fifteen times too long, and its response at 1 THz exists only if the potential drop is confined to the neck. A statement that geometric diodes are or are not terahertz rectifiers has no content without the length, and the dc asymmetry of a long device is no guide: the best dc asymmetry at $\lambda = 700$ nm in Table II belongs to devices between 0.55 and 2 μm, and only the shortest of them still responds at 1 THz in a uniform field.

Length is necessary and not sufficient. The 0.55 μm funnel meets the criterion, and with an imposed potential it appears to give 0.7 A/W per absorbed watt whatever the profile. With the potential computed it gives 0.12 A/W at the room-temperature mean free path, because the bias does not go where either imposed profile

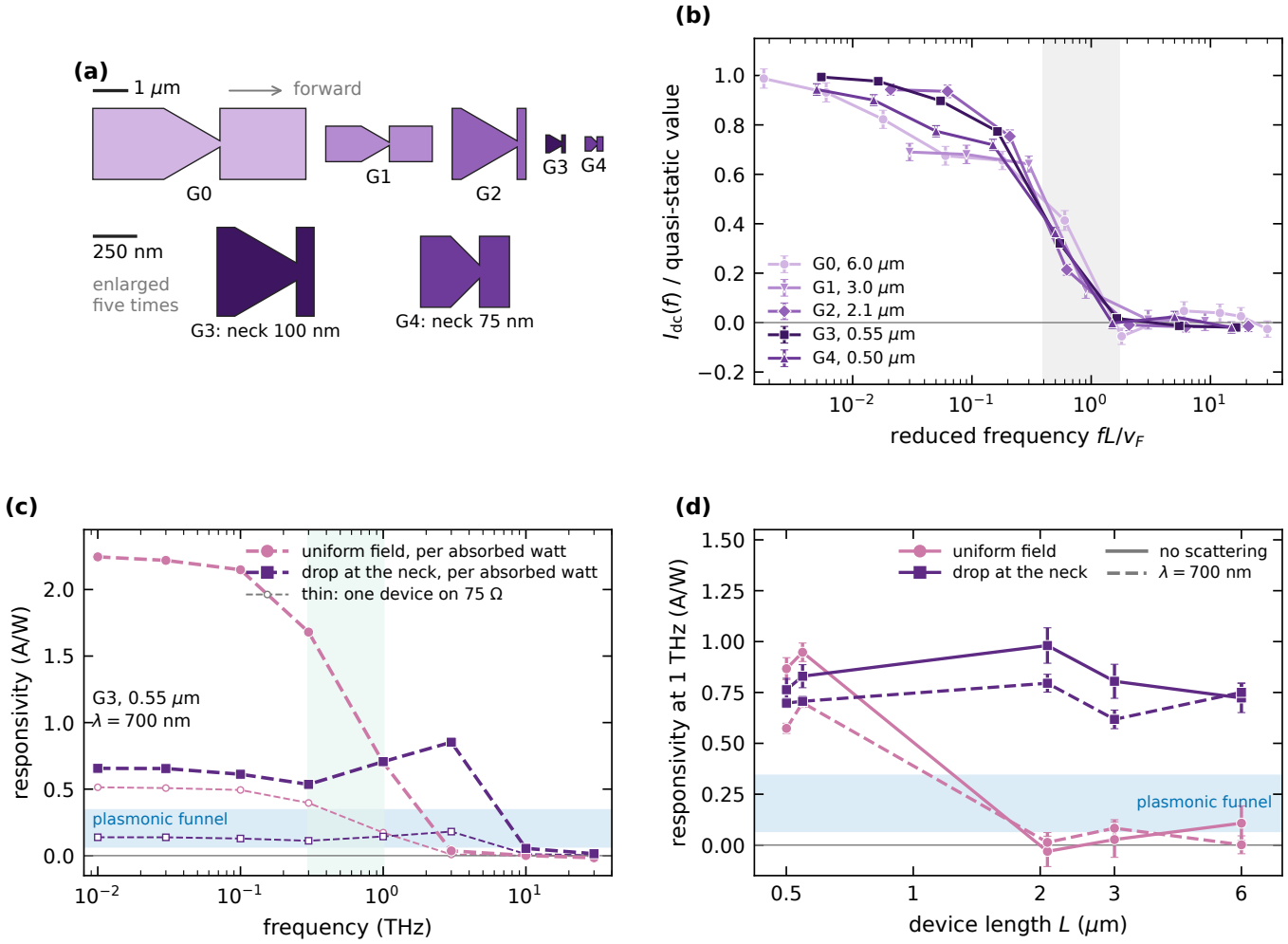


FIG. 9. Length scaling of the ballistic response at an ac amplitude of 0.1 V. (a) The five funnels of Table II to scale, and the two short ones enlarged. (b) Rectified current relative to its quasi-static value against the reduced frequency, for a uniform field and no scattering; the band marks the fall from half response to zero. (c) Responsivity of the 0.55 μm funnel G3 at $\lambda = 700$ nm: per absorbed watt, Eq. (13) (thick), and for one device on a 75 Ω antenna (thin, open symbols). (d) Responsivity per absorbed watt at 1 THz against device length for the two field profiles. The blue band in (c) and (d) is the range of the plasmonic funnel of Sec. IV, 0.07 to 0.34 A/W. Error bars are one standard error.

puts it: a quarter is spent at the contacts and the rest is centred on the neck. This is a caution beyond the present device. A uniform field is a common assumption of simulations of geometric diodes; self-consistent electrostatics was already known to roughly halve the static asymmetry [35], and here it reduces a static asymmetry of 1.56 to 1.05 and the terahertz response sixfold. Three further conditions follow from Sec. V E. The mean free path is the least of them: from no scattering to 100 nm the response per absorbed watt changes from 0.13 to 0.10 A/W. The edges are the strictest: with half of the wall reflections diffuse the response is lost. And the gate must be far: under a gate at 30 nm a straight strip with its drain tied to the gate rectifies twice the current of the funnel shape at 1 THz, so that the circuit takes over in the ballistic gas as it did in the fluid. Shape-based rectification at terahertz frequencies, then, exists in one of the three transport limits, in devices of half a micrometre and below with specular edges and a distant gate, and at its best it equals the circuit term of the plasmonic element at room temperature.

B. What the elements are good for

Table IV and Fig. 11(b) place the simulated elements against measured graphene terahertz detectors. At room temperature the best reported devices have noise-equivalent powers of 100 to 130 pW/Hz$^{1/2}$ referred to the optical power at 0.4 THz [62, 63], with an estimate of 51 pW/Hz$^{1/2}$ at 0.33 THz for an epitaxial bilayer device [64], and 80 to 160 pW/Hz$^{1/2}$ referred to a diffraction-limited spot at 2.5 to 2.8 THz [65, 66]; most others lie between 0.5 and 2 nW/Hz$^{1/2}$ [67–70], the first

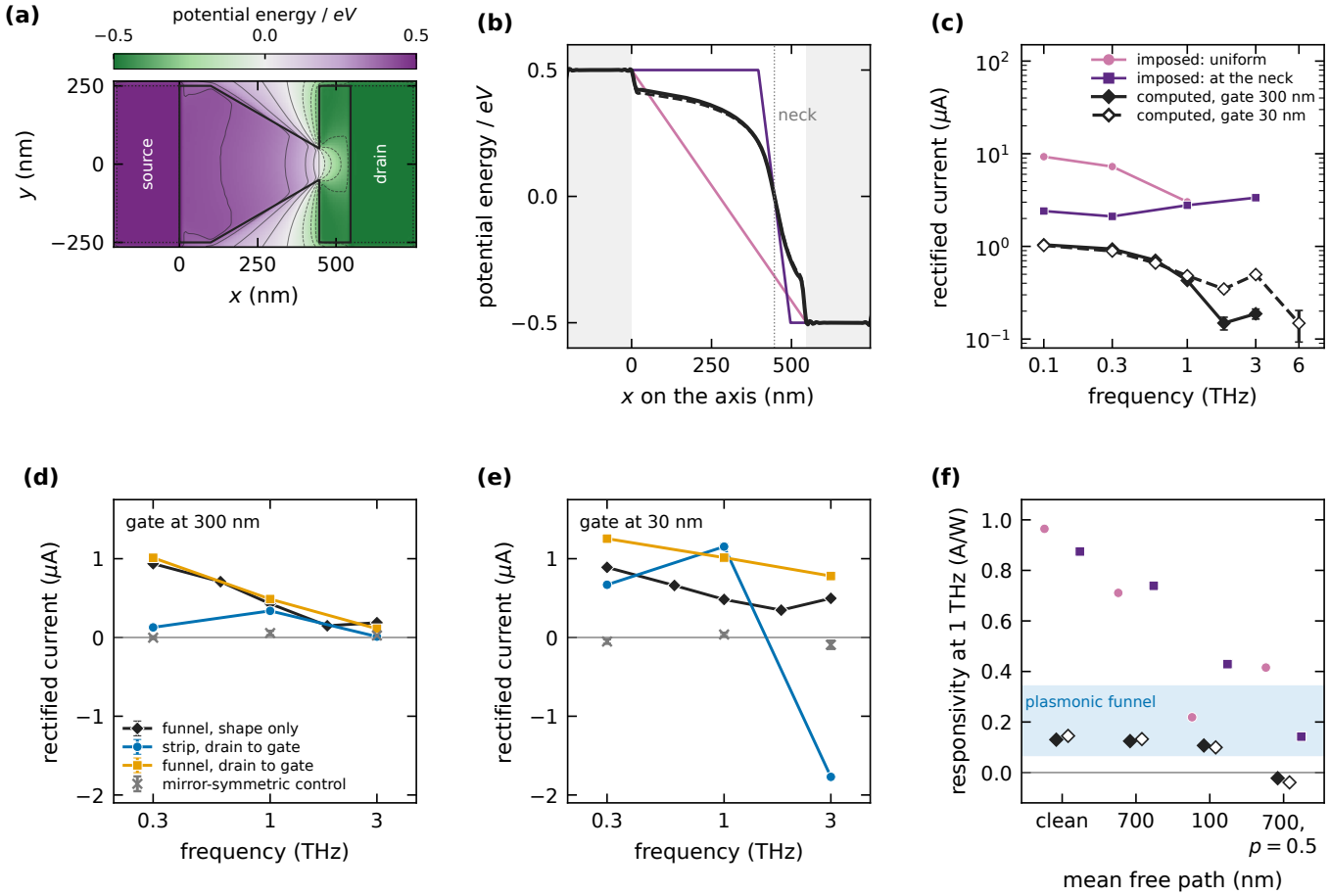


FIG. 10. The 0.55 μm funnel G3 with the field computed; $\lambda = 700$ nm and an amplitude of 0.1 V except in (f). (a) Static potential energy at a bias of 0.1 V with the gate at 300 nm, in units of the bias; contours every 0.1; the dotted rectangles are the contact strips. (b) The same along the axis for the two gate distances, against the two imposed profiles of Sec. V D; colours and line styles as in (c). (c) Rectified current against frequency under the drive that leaves only the shape, for the imposed fields and for the computed field with the gate at 300 nm (full symbols) and 30 nm (open symbols); points within two standard errors of zero are not drawn. (d), (e) Shape against circuit for the two gate distances: the funnel under the shape-only drive, a straight strip of the same length with its drain tied to the gate, the funnel with its drain tied to the gate, and the mirror-symmetric funnel under the shape-only drive, which must give zero. (f) Responsivity at 1 THz for no scattering, mean free paths of 700 and 100 nm, and half-diffuse walls: per absorbed watt for the computed field (diamonds, full and open as in (c)), and by Eq. (13) for the imposed fields (circles: uniform; squares: drop at the neck). The band is the range of the plasmonic funnel. Error bars are one standard error.

devices were at tens of nW/Hz$^{1/2}$ [27], and four-terminal ballistic rectifiers of encapsulated graphene have been reported at 34 pW/Hz$^{1/2}$ below 0.7 THz [7] and, at 3 THz and 4 K, at 20 to 30 pW/Hz$^{1/2}$ [8]; funnel rectifiers of chemical vapor deposited graphene, with a mean free path below the device size, are at 29 nW/Hz$^{1/2}$ [9]. The Johnson-limited 60 pW/Hz$^{1/2}$ of the DS-terminated 1 μm funnel at 300 K, or 150 pW/Hz$^{1/2}$ with 300 $\Omega\,\mu$m contacts, is therefore where the best measured devices already are, and its voltage responsivity of 50 V/W is within their range of 1 to 100 V/W. The idealised model holds no hidden order of magnitude; the comparison supports the model rather than promising an improvement. The matched array of 0.55 μm ballistic funnels is at 130 pW/Hz$^{1/2}$ at 1 THz and room temperature, twice the figure of the plasmonic element at that temperature and again where the best measured room-temperature devices are; it assumes nineteen devices matched to the antenna, specular edges and no parasitic capacitance, and it is four times the value reported for the encapsulated ballistic rectifiers [7]. At low temperature, encapsulated devices reach 0.2 pW/Hz$^{1/2}$ at 10 K and 0.13 THz [28, 71] and 18 pW/Hz$^{1/2}$ at 77 K and 2.8 THz [66]; the simulated 17 to 27 pW/Hz$^{1/2}$ at 100 K, and 1 pW/Hz$^{1/2}$ at low density, lie in the same range. Three caveats apply. Nearly all reported values, like ours, are computed from the Johnson noise of the measured resistance and not from measured noise. The reference power differs between reports (total beam, power delivered to the antenna, or power scaled to a diffraction-limited spot),

TABLE III. Rectification by geometry and regime. A: ballistic asymmetry at 0.1 V; D: asymmetry of the voltage in steady flow; R_I: current responsivity into a 75 Ω source; R_{abs}: per absorbed watt, Eq. (13) for the imposed fields and Eq. (15) for the computed one. "Shape only" is the symmetric termination; DS has the drain tied to the gate.

Regime	Funnel	Classical Tesla valve	What limits it
Ballistic, dc	$A = 1.6$ to 1.9 clean; 1.1 to 1.6 at $\lambda = 700$ nm, by length and field profile	$A = 0.89$ to 1.08, no systematic sign	reciprocity; needs the bias landscape
Viscous, dc, no gate	$\lvert D-1\rvert \le 10^{-2}$ within the drift limit	10^{-3} to 3×10^{-2} within the drift limit	drift speed below $0.3\,v_F$ caps Re
Viscous, dc, gated	contacts exchanged: $< 10^{-3}$; polarity reversed: $0.48\,V/U_0$	the same polarity term; exchange $< 10^{-3}$	the polarity term is the gate, in any channel
Plasmonic, shape only	10^{-4} to 0.13 A/W; 0.22 A/W at taper resonance	10^{-5} to 10^{-3} A/W; 0.02 A/W at resonance	plasmon damping; saturates first
Plasmonic, DS	0.07 to 0.34 A/W; $\eta < 1\%$	0.02 to 0.1 A/W at 6 to 11 kΩ	r^2; gate overdrive; contacts
Ballistic, 1 THz, 6 μm	$R_I = 0.05$ A/W (uniform field); 0.2 to 0.45 A/W (drop at neck)	none ($A = 1$ at dc)	length: $fL/v_F = 6$
Ballistic, 1 THz, 0.55 μm, imposed field	$R_{\mathrm{abs}} = 0.7$ A/W, either profile; 0.14 to 0.42 A/W with rough edges or $\lambda = 100$ nm	none	the field is not computed
Ballistic, 1 THz, 0.55 μm, computed field	$R_{\mathrm{abs}} = 0.12$ A/W; 0.10 A/W at $\lambda = 100$ nm; $\lvert R_{\mathrm{abs}}\rvert \le 0.04$ A/W with half-diffuse edges; $R_I = 0.012$ to 0.026 A/W	not computed	contact drops; edges; under a 30 nm gate the circuit term is larger; $\eta \sim 10^{-4}$

and values normalised to the active device area are not comparable and are left out. And part of the measured response of graphene detectors is photothermoelectric [63, 65, 72], which the model does not contain. The room-temperature incumbent is better than any of these: commercial zero-bias Schottky detectors have typical noise-equivalent powers of 5 to 27 pW/Hz$^{1/2}$ from 0.22 to 1.1 THz [73], which the simulated plasmonic funnel reaches only at 100 K and the simulated ballistic array does not reach.

For energy conversion the conclusion does not depend on such comparisons, and its framework is not new. The efficiency of a field-effect detector is quadratic in the amplitude up to the gate swing and has its maximum, of the order of ten per cent, below threshold [32]; silicon transistor rectennas work there and have reached 13.6% at 263 GHz [33]. What the present calculation adds is the graphene case. A gapless monolayer has no subthreshold regime, the voltage scale of its gated fluid is the gate overdrive, and the efficiency at an amplitude equal to the overdrive is the pure number $r^2 R_{\mathrm{ant}}/R_{\mathrm{dc}}$ of Eq. (11). For a funnel on a real antenna that number is of order 10^{-2}, and no density or gate distance in the hydrodynamic window raises it; a low density or a thin gate moves the power at which the bound is reached from a milliwatt to a microwatt, which is useful for matching a detector's dynamic range to its signal, and nothing more. Realistic contacts of 100 to 300 Ω μm cost a further factor two to five. The ballistic funnel has a lower voltage scale, the Fermi energy, and a lower efficiency still: 10^{-4} at 13 μW per device, the largest amplitude computed. A conversion efficiency of 58% at 0.9 THz has been predicted by simulation for a graphene rectenna [74], but its rectifier is a Schottky junction between graphene and a semiconductor, not a geometric or plasmonic element. Within the mechanisms studied here a graphene rectenna has no harvesting regime.

A designer who wants a graphene terahertz detector from these shapes should therefore read the results as follows. In a gated device of a micrometre or more, use the funnel as a matched horn, not as a diode: its value is the impedance it presents and the taper resonance. Terminate the channel asymmetrically, since that is where the rectification comes from, put the saturation knee at the expected power by choice of density and gate distance, and spend the fabrication effort on the contacts, because a resistive drain contact removes the asymmetry it is supposed to serve. If the device is to rectify by its shape, make it short, by Eq. (16): half a micrometre for 1 THz, leads included. Spend the effort on the edges before the mobility, since a mean free path of 100 nm costs a fifth of the response and half-diffuse edges cost all of it; keep the gate farther away than the device is long, or accept that the gate connection does the rectifying; and provide for the impedance, since one such funnel is a kilohm. Expect no more than the circuit term gives. Do not judge a geometric diode for terahertz use by its dc asymmetry, nor by a simulation with an imposed field. Do not add Tesla loops.

C. Limitations

There is no experiment here: the solvers are validated against analytic limits and one measured number, and the agreement with that number depends on the field profile and on the mean free path at the level of a factor of several (Sec. III). Most idealisations favour the device.

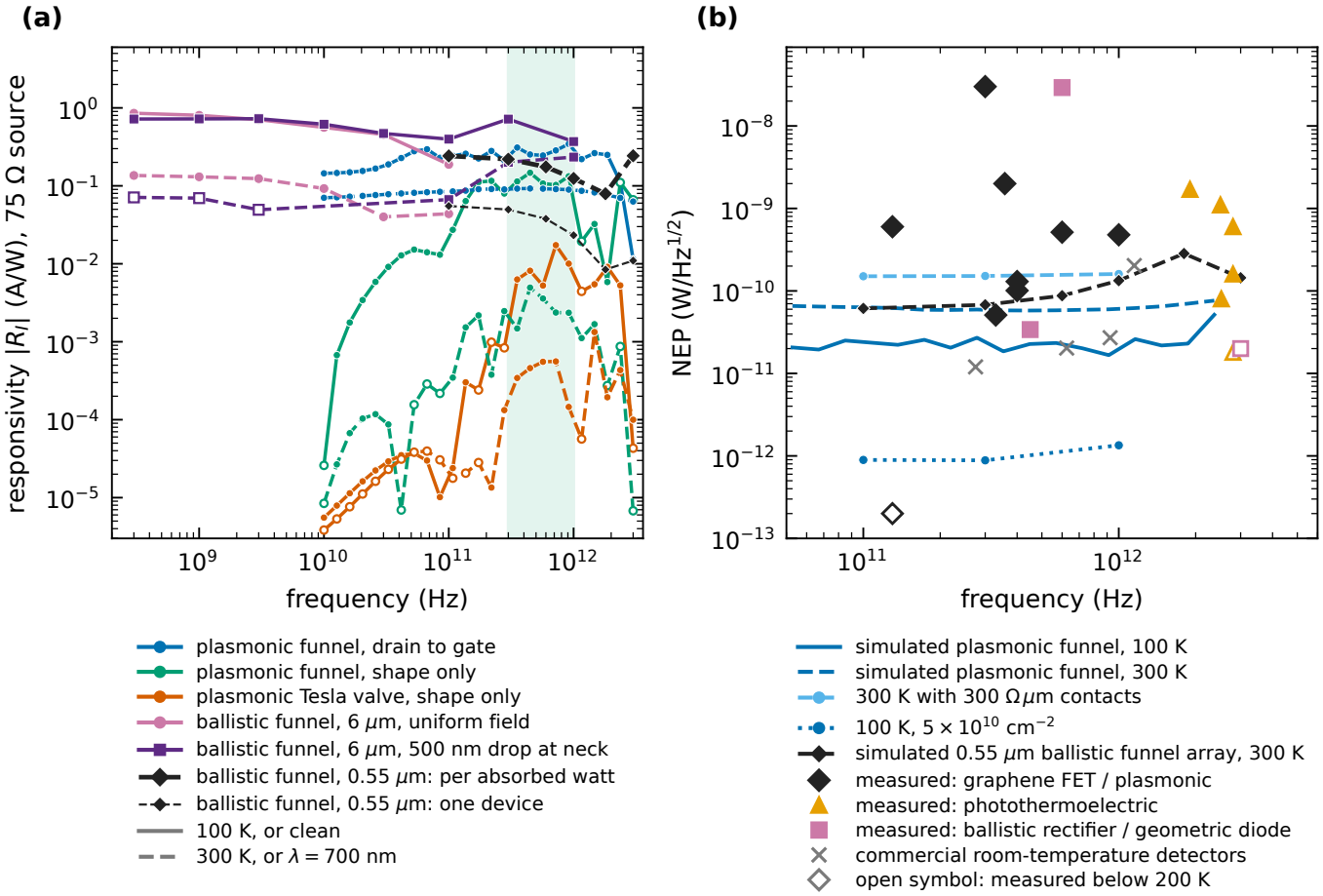


FIG. 11. Synthesis. (a) Current responsivity against frequency for every mechanism: the plasmonic 1 μm funnel with the Dyakonov-Shur termination and with the symmetric drive that isolates the shape, the Tesla valve, the 6 μm ballistic funnel with a uniform field and with the potential drop confined to the neck, all into a 75 Ω source, and the 0.55 μm ballistic funnel with the field computed (gate at 300 nm, shape-only drive), for one device on the same source and per absorbed watt. Solid lines: 100 K parameters or the clean ballistic limit; dashed: 300 K or $\lambda = 700$ nm. Points consistent with zero are not drawn. (b) Noise-equivalent power of the simulated elements, which is Johnson limited and assumes ideal coupling, against measured graphene terahertz detectors (Table IV); the ballistic array is that of Sec. V E.

That is the safe direction for the negative results, on the Tesla valve and on the conversion efficiency, and the unsafe one for the positive result on the short ballistic funnel, which computing the field reduced sixfold and which remains an upper estimate.

The ballistic model of the 6 μm device and of the length scaling imposes its potential and has no displacement current. For the 6 μm device the terahertz result depends entirely on the length of the drop, and the scaling curve of Fig. 9(b) was confirmed with a computed field for one device only, through its roll-off. For the 0.55 μm funnel the field is computed, in a model with limits of its own. The dielectric is uniform. The contacts are ideal equipotential strips that absorb and inject without resistance and without doping the graphene beneath them; a quarter of the bias drops at the contact planes within three grid cells, and since the rectified current goes roughly as the square of what remains at the neck, a different contact would change the result. The electrons are classical particles in a classical field, with five to seven modes in the neck, and they interact through the common field only; short-range electron-electron scattering enters through the 100 nm bound. Two wall specularities were computed, and the response is lost between them. The antenna is a voltage source, with its 75 Ω entering through the computed admittance; the matched array is an idealisation, and no array or antenna was modelled. Rectified currents below about 0.05 μA are not resolved. Two results of Sec. V E are reported without an explanation: the reversal of the strip's current near the plasma resonance at 3 THz, where three frequencies do not resolve the resonance, and the rise of the funnel's current above the square law at 0.2 V. No model of the crossover between the ballistic and the fluid limit is offered: the two terahertz models describe the same gate and nearly the same density, 1.4×10^{12} and 10^{12} cm^{-2}, in opposite

TABLE IV. Measured graphene terahertz detectors and the simulated element. Responsivity and NEP as stated by the authors; the last column gives the power to which they are referred: the power in a diffraction-limited spot, the full optical beam power, the power delivered to the antenna, an extrinsic figure, the waveguide input, or, for this work, the available power with ideal coupling or the power absorbed by a matched array. NEPs are estimated from Johnson noise except the one marked with a double dagger, which was measured; the commercial values are typical specifications. Entries marked with a dagger were read from the abstract only.

Reference	Device and mechanism	f (THz)	T (K)	Responsivity	NEP (pW/Hz$^{1/2}$)	Power
[27]	FET, overdamped plasma wave	0.3	300	0.10 to 0.15 V/W	3×10^4	spot
[67]	bilayer FET	0.29 to 0.36	300	1.2 V/W	2×10^3	spot
[68]†	CVD FET, bow-tie antenna	0.6	300	14 V/W	515	optical
[62]	CVD FET, bow-tie antenna	0.4	300	74 V/W	130	optical
[63]	CVD FET, bow-tie antenna	0.4	300	59 to 63 V/W	101 to 114	optical
[69]	encapsulated FET	0.13	300	20 V/W	600	antenna
[28]	encapsulated bilayer FET	0.13	10	> 3 kV/W	0.2	antenna
[70]	thermoelectric, patch antenna	1.9	300	4.9 V/W	1.7×10^3 ‡	optical
[65]	photothermoelectric p-n junction	2.5	300	105 V/W	80	spot
[66]	photothermoelectric and plasma wave	2.8	300; 77	49; 180 V/W	160; 18	spot
[7]†	four-terminal ballistic rectifier	up to 0.69	300	764 V/W	34	extrinsic
[64]†	epitaxial bilayer FET	0.33	300	30 V/W	51	not verified
[8]	four-terminal ballistic rectifier	3.0	4; 300	77; 6 V/W	20 to 30 (4 K)	spot
[9]	CVD four-terminal funnel rectifier	0.49 to 0.68	300	0.77 V/W	2.9×10^4	optical
[73]	commercial zero-bias Schottky	0.22 to 1.1	300	750 to 2200 V/W	5 to 27	waveguide
This work	1 μm funnel, DS, simulated	0.3 to 1	300	0.09 A/W, 52 V/W	60; 150 with contacts	available
This work	same	0.3 to 1	100	0.21 to 0.34 A/W, 36 to 58 V/W	17 to 27	available
This work	same, 5×10^{10} cm^{-2}	0.3	100		0.9	available
This work	0.55 μm ballistic funnel, computed field, matched array, simulated	1	300	0.12 A/W	130	absorbed

limits of the electron-electron scattering rate, and a real device at room temperature lies between them.

The plasmonic and the gated steady-flow models use a local gate capacitance, which is least accurate for the 175 nm neck under a 30 nm gate; they have ideal contacts except where stated, perfect coupling to an antenna represented by its resistance, and Johnson noise only. They contain no electron heating, and so none of the photothermoelectric response that contributes to measured graphene detectors; the fluid description is quantitative to about 0.3 THz at 100 K and 1 THz at 300 K, where $\omega\tau_{ee} = 1$, and indicative beyond. The low-bias shape term of the gated steady flow is bounded, not resolved. Finally, two experiments on electronic Tesla valves are outside what the models reproduce: the rectification reported in graphene at low current [17] is not obtained from steady hydrodynamics with or without the gate, and the large diodicity above threshold in GaAs [19] belongs to a regime of unsteady, heated flow that was not computed.

VII. CONCLUSION

We have computed the rectification of graphene funnels and of an electronic Tesla valve in the ballistic, viscous and plasmonic transport limits, from dc to 30 THz and from the small-signal limit to saturation, and separated in each what the shape does from what the circuit does. The Tesla valve does not rectify ballistically, rectifies in the viscous liquid only beyond the drift velocity the graphene Fermi liquid allows, and is an order of magnitude below the funnel at terahertz frequencies. In the electron fluid the funnel's rectification belongs to

the circuit: a steady current in a gated channel of any shape shows an asymmetry of 0.48 times the bias over the gate overdrive under reversal of polarity and less than 10^{-3} under exchange of the contacts, and the terahertz response, 0.07 to 0.34 A/W into a 75 Ω antenna, is the Dyakonov-Shur asymmetry of the terminations, the shape contributing as a matched horn near a taper resonance. Its conversion efficiency follows the square law of field-effect detectors up to a saturation power that scales as the square of the gate overdrive and peaks below one per cent at every density and gate distance; contact resistance attacks the asymmetry itself.

In the ballistic limit the shape rectifies, and the frequency to which it does so is set by the length of the device: the response is a function of fL/v_F, half at 0.4 and gone at 1.7 for a field spread over the device. The 6 μm funnel that anchors the dc literature is too long by a factor of fifteen at 1 THz. A 0.55 μm funnel is short enough, and with an imposed field it gives 0.7 A/W per absorbed watt at 1 THz for either of two profiles. With the field computed from the charge of the electrons, a gate and the contacts it gives 0.12 A/W at the room-temperature mean free path and 0.10 A/W at 100 nm, because the bias drops partly at the contacts and otherwise symmetrically about the neck. With half-diffuse edges the response is lost, and under a gate at 30 nm the gate connection of a straight channel rectifies more than the funnel shape.

Geometry, then, can make a graphene terahertz rectifier, in one regime, at one size and with specular edges, and what it makes there is no better than what the circuit makes. It does not make a rectenna that harvests: the efficiency is below one per cent in every case and of order 10^{-4} for the ballistic funnel. What it makes is a square-law detector with a Johnson-limited noise-equivalent power of 60 to 130 pW/Hz$^{1/2}$ at room temperature, for the plasmonic element if contacts of 100 Ω μm and the assumed scattering times can be had, and for the ballistic one if a half-micron funnel with specular edges can be matched to an antenna. Three things would test or change these statements and are outside the models used here: a measurement of the dc asymmetry and the terahertz response of one short funnel against the distance of its gate and the quality of its edges; a rectification mechanism that is neither ballistic, viscous nor plasmonic, of which the photothermoelectric effect is the obvious example; and a material whose electron liquid sustains a higher drift velocity relative to its plasma and Fermi velocities.

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Appendix A: Second-order plasmonic response

In units of the contact width W, the plasma speed s_0 and the gated density n_0, Eqs. (5) and (6) contain the friction $\Gamma = W/s_0\tau_{\rm mr}$, the plasma Reynolds number $\mathrm{Re}_s = s_0W/\nu$, the collision time $s_0\tau_{ee}/W$ and the quantum-capacitance fraction q, with the closure $h(n) = 2(1-q)(\sqrt{n}-1)+q\ln n$ and $h''(1) = -(1+q)/2 \equiv \kappa$. Writing $n = 1 + \mathrm{Re}(n_1 e^{i\Omega t}) + n_{20}$ and $\boldsymbol{u} = \mathrm{Re}(\boldsymbol{u}_1 e^{i\Omega t}) + \boldsymbol{u}_{20}$, the first order is

$$i\Omega n_1 + \nabla\cdot\boldsymbol{u}_1 = 0, \tag{A1}$$

$$(i\Omega+\Gamma)\boldsymbol{u}_1 + \nabla n_1 - \nu(\Omega)\nabla^2\boldsymbol{u}_1 = 0, \tag{A2}$$

with $\nu(\Omega) = \mathrm{Re}_s^{-1}/(1+i\Omega s_0\tau_{ee}/W)$, and the time average of the second order is

$$\nabla\cdot\boldsymbol{u}_{20} = -\tfrac{1}{2}\mathrm{Re}\,\nabla\cdot(n_1\boldsymbol{u}_1^*), \tag{A3}$$

$$\Gamma\boldsymbol{u}_{20} + \nabla P - \mathrm{Re}_s^{-1}\nabla^2\boldsymbol{u}_{20} = -\tfrac{1}{2}\mathrm{Re}\,(\boldsymbol{u}_1^*\cdot\nabla)\boldsymbol{u}_1, \tag{A4}$$

where $P = n_{20} + (\kappa/4)|n_1|^2$. The dc contact voltage is $P + c_V|n_1|^2$ with $c_V = 1/8$ for the Dirac closure and zero for shallow water, and the dc particle current through a contact is the flux of $\boldsymbol{u}_{20} + \mathrm{Re}(n_1\boldsymbol{u}_1^*)/2$. Each contact is an equipotential with one unknown voltage per order, closed by Kirchhoff's law for its termination: $RI_c + V_c = V_{a,c}$ for a source of resistance R, $I_c = 0$ for an open contact. The same structure with a matched load at dc gives the rectified power. Voltages are in units of $V_u = m^*s_0^2/e$, resistances in units of m^*s_0/e^2n_0W (65 Ω for $W = 4\,\mu$m), and frequency in units of $s_0/2\pi W$ (0.142 THz).

A contact resistance R_c enters as a change of the ports only: the source becomes a Thevenin source of resistance $R_{\rm ant} + R_c$ and the drain one of zero amplitude and resistance R_c. In second order the dc problem is linear, so the short-circuit current and open-circuit voltage satisfy $I_{\rm sc}(R_{\rm dc} + 2R_c) = V_{\rm oc}$ exactly, which is one of the ladder tests. At the signal frequency the antenna does not see the device impedance plus $2R_c$, because the source and drain currents differ by the gate displacement current.

Appendix B: Saturation estimate on the density and gate grid

The peak efficiencies of Fig. 7(b) are $\eta_{\rm peak} = 2r^2(R_{\rm ant}/R_{\rm dc})\,(0.7)^2\,\mathcal{S}$, with r and $R_{\rm dc}$ from the small-signal solver at each grid point and $\mathcal{S}$ the ratio of the time-domain efficiency to the square law at $V_a = 0.7\,V_u$ for the reference device (0.86, 0.89, 0.60 at 100 K and 1.14, 1.10, 0.81 at 300 K for 0.1, 0.3, 1 THz). The assumption that saturation occurs at a fixed V_a/V_u was

tested by a time-domain ladder at $(5\times10^{11}\,\mathrm{cm}^{-2}, 10\,\mathrm{nm})$: $\mathcal{S}$ agrees to 0.2% at 300 K and is 0.76 instead of 0.89 at 100 K, so the 100 K peak values at low density and thin gates are upper estimates by 10 to 20%.

Appendix C: Steady gated flow in one dimension

In a gradual channel of width $W(x)$ carrying the current I the continuity equation gives $nuW = I$. Without the convective and viscous terms the steady form of Eq. (6) is $\Gamma u = -h'(n)\,dn/dx$ in the units of Appendix A, so that

$$n\,h'(n)\,\frac{dn}{dx} = -\frac{\Gamma I}{W(x)}. \tag{C1}$$

With $G'(n) = nh'(n)$, i.e. $G = n^2/2$ for shallow water and $G = \frac{2}{3}(1-q)n^{3/2} + qn$ for the Dirac closure, integration from the driven contact, where $n = n_s$, to the contact tied to the gate, where $n = 1$, gives Eq. (9). The voltage of the driven contact is $\Phi(n_s)$ with $\Phi = (1-q)(n-1) + 2q(\sqrt{n}-1)$ in units of V_u. The width enters only through $\int dx/W$, which is unchanged when the channel is turned end for end, so exchanging the contacts cannot produce an asymmetry. Reversing the current changes the sign of the right-hand side; expanding to second order with $G''(1) = (1-q)/2$ and $\Phi''(1) = -q/2$,

$$V = \rho I - \tfrac{1}{4}(\rho I)^2, \qquad \rho = \Gamma \int_0^L \frac{dx}{W}, \tag{C2}$$

and the ratio of the two voltage magnitudes is $1+\rho|I|/2 = 1 + V/2V_u$, which is Eq. (10) with $U_0 = (1-q)V_u$ and $q = 0.039$; for shallow water the coefficient of $(\rho I)^2$ is $\frac{1}{2}$ and the ratio is $1 + V/U_0$. In a uniform channel the convective term can be kept: the equation becomes $(nh' - j^2/n^2)\,dn/dx = -\Gamma j$ with $j = I/W$, with the integral $G(n) + j^2/n$, and the two-dimensional solver reproduces this closed form to 1.1×10^{-4} for both closures and both signs up to $V = 0.3\,U_0$. In a channel of varying width the convective term brings in dW/dx, and with it the inertial shape term of Sec. III, which is second order in the current.

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